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Phase equilibria and thermodynamic properties in the RE-Ni (RE = rare earth metals) binary systems

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ABSTRACT

The RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems have been optimized thermodynamically using the CALPHAD (CALculation of PhAse Diagram) method based on experimental phase equilibrium and thermodynamic property data. The solution phases including the liquid phase and terminal solid solution phases were modeled by the substitutional solution model with the excess Gibbs energy being described by the Redlich–Kister polynomials. The binary intermetallic compounds RE_aNi_b were treated either as stoichiometric phases or by using the two-sublattice model to consider the homogeneity ranges of the compounds. A set of self-consistent thermodynamic parameters of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems was obtained, which can be used to reproduce well the experimental data. Furthermore, in combination with the reported assessments of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary systems, the phase equilibria and thermodynamic properties of the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems were examined systematically. A trend was shown for the variation of phase equilibria and thermodynamic properties as a function of the RE atomic number. In general, as the atomic number of the rare earth metals increases, the phase transition/reaction temperatures of the RE-Ni intermetallic compounds become higher and higher, and the enthalpies of mixing of liquid RE-Ni (except for Sm-Ni) alloys as well as the enthalpies of formation of the RE-Ni (except for Y-Ni and Sm-Ni) intermetallic compounds become increasingly negative. This trend demonstrates stronger RE-Ni bonds in binary systems containing heavier rare earth elements.

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Introduction

The intermetallic compounds formed by rare earth metals (RE) and 3d transition metals (TM = Fe, Co, Ni and Mn) exhibit special physical and chemical properties and have been used in industrial applications mainly as functional materials, such as permanent magnets, magnetic refrigeration materials, hydrogen storage materials, catalysis materials and superconducting materials [1–4]. In particular, the intermetallic compounds formed by rare earth metals and Ni show excellent hydrogen storage properties and magnetic performances [1, 5]. For example, $LaNi_5$ compound has been produced commercially as the anode materials in the field of nickel-metal hydride (Ni-MH) batteries due to its characteristics of easy activation, high storage capacity and long cycle life [6–16]; La_2Ni_7 compound shows good electrochemical properties [17–19]. RE-Ni-Si/In intermetallic compounds find profound applications in the magnetic refrigeration technology [20–22]. RENi and $RENi_2$ intermetallic compounds are excellent low-temperature magnetocaloric materials, and they are of great significance to the development of low-temperature magnetic refrigeration technology [23–27]. In order to synthesize and explore these novel functional materials, it is essential to have an accurate understanding of the phase equilibria and thermodynamic properties of the RE-Ni-based alloy systems.

The phase equilibria and thermodynamic properties of the key RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems have been studied extensively through experimental investigation and thermodynamic calculation, but many inconsistencies and disagreements still exist among the reported results and remain to be critically examined. For example, the temperatures of invariant reactions determined experimentally by Fartushna et al. [28] in the La-Ni binary phase diagram are obviously different from the calculated results [29–32], and the La-Ni binary phase diagrams were modified by Fartushna et al. [28]. Mardani et al. [33] found a new intermetallic compound Ce_5Ni_{19} experimentally and modified the Ce-Ni binary phase diagrams accordingly. Meanwhile, the enthalpy of mixing of liquid Nd-Ni alloys calculated by Hussain et al. [34], Huang et al. [35], Luo et al. [36], Du and Clavaguera [37] is not consistent with the experimental data measured by Sudavtsova et al. [38], and the phase equilibria of the Tb-Ni binary system need to be re-optimized according to the experimental results determined by Yao et al. [39]. In addition, the enthalpy of mixing of liquid Ho-Ni alloys calculated by Kardelass et al. [40] is too positive and breaks the systematic trend of thermodynamic properties in the RE-Ni binary systems. Therefore, a re-assessment of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems was performed in this work using the CALPHAD (CALculation of PhAse Diagram) method based on available experimental data. The phase equilibria and thermodynamic properties of the RE-Ni binary systems were discussed systematically by combining with the reported thermodynamic calculation of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary systems. A complete set of thermodynamic descriptions of the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems was obtained finally, which provides a good basis for the development of a thermodynamic database of multicomponent alloy systems including rare earth metals (RE) and 3d transition metals.

Thermodynamic calculation

Thermodynamic models

In the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems, the molar Gibbs energy of a solution phase φ (e.g., liquid, fcc-Ni, bcc-La, fcc-La, dhcp-La, bcc-Ce, fcc-Ce, bcc-Nd, dhcp-Nd, bcc-Tb, hcp-Tb, bcc-Ho and hcp-Ho) is expressed by the substitutional solution model as follows:

$$G_m^{\varphi} = {}^0G_{\mathrm{Ni}}^{\varphi}x_{\mathrm{Ni}} + {}^0G_{\mathrm{RE}}^{\varphi}x_{\mathrm{RE}} + RT\sum_{i=\mathrm{RE,Ni}} x_i \ln x_i + x_{\mathrm{RE}}x_{\mathrm{Ni}}\sum_{n=0} {}^nL_{\mathrm{RE,Ni}}^{\varphi}(x_{\mathrm{RE}} - x_{\mathrm{Ni}})^n + {}^{\mathrm{mag}}G_m^{\varphi} \quad (1)$$

$${}^nL_{\mathrm{RE,Ni}}^{\varphi} = A_n + B_nT \quad (2)$$

$${}^{\mathrm{mag}}G_m^{\varphi} = RT\ln(\beta_0 + 1)g(\tau) \quad (3)$$

where ${}^0G_{\mathrm{Ni}}^{\varphi}$ and ${}^0G_{\mathrm{RE}}^{\varphi}$ are the molar Gibbs energy of the pure element Ni and RE, respectively, in the structure of the phase φ with respect to the standard reference state at 298.15 K and 1 bar, and their descriptions are taken from the SGTE (Scientific Group Thermodata Europe) database compiled by Dinsdale [41]. x_{Ni} and x_{RE} are the mole fractions of the element Ni and RE, respectively. R is the universal

gas constant, and T is the absolute temperature in Kelvin. ${}^{n}L^{\varphi}_{\mathrm{RE,Ni}}$ is the n-th-order interaction parameters as expressed by two constants A_n and B_n to be optimized. ${}^{\mathrm{mag}}G^{\varphi}_{m}$ is the magnetic contribution to the molar Gibbs energy of the phase φ. β_0 is the Bohr magneton number. τ is the normalized temperature as expressed by $\tau = T/T^{\varphi}_{\mathrm{C}}$, where T^{φ}_{C} is the Curie temperature of the phase φ. According to the equation suggested by Hillert and Jarl [42], $g(\tau)$ is expressed by:

$$g(\tau) = \begin{cases} 1 - \frac{1}{M}\left[\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p}-1\right)\left(\frac{\tau^3}{6}+\frac{\tau^9}{135}+\frac{\tau^{15}}{600}\right)\right] (\tau > 1) \\ -\frac{1}{M}\left[\frac{\tau^{-5}}{10}+\frac{\tau^{-15}}{315}+\frac{\tau^{-25}}{1500}\right] (\tau < 1) \end{cases} \tag{4}$$

$$\mathrm{M} = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p}-1\right) \tag{5}$$

in which p is determined by the structure of the phase φ (0.4 for bcc phase and 0.28 for other phases).

In the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems, the following intermetallic compounds are regarded as stoichiometric compounds due to the lack of the experimental information and/or their narrow homogeneity ranges: La_3Ni, La_7Ni_3, LaNi, La_2Ni_3, La_7Ni_{16}, $LaNi_3$, La_2Ni_7, La_5Ni_{19}, $LaNi_5$, Ce_7Ni_3, CeNi, $CeNi_3$, Ce_2Ni_7, Ce_5Ni_{19}, Nd_3Ni, Nd_7Ni_3, NdNi, $NdNi_2$, $NdNi_3$, Nd_2Ni_7, $NdNi_5$, Nd_2Ni_{17}, Tb_3Ni, Tb_3Ni_2, TbNi, $TbNi_2$, $TbNi_3$, Tb_2Ni_7, $TbNi_5$, Tb_2Ni_{17}, Ho_3Ni, Ho_3Ni_2_LT, Ho_3Ni_2_HT, HoNi, $HoNi_3$, Ho_2Ni_7, $HoNi_5$ and Ho_2Ni_{17}. Because the Curie temperatures of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary compounds are much lower than 298 K, the magnetic contribution to the Gibbs energy of these intermetallic compounds $\mathrm{RE}_a\mathrm{Ni}_b$ was ignored in this work. The molar Gibbs energies of these intermetallic compounds $\mathrm{RE}_a\mathrm{Ni}_b$ in the RE-Ni binary systems are expressed by using the Neumann–Kopp rule:

$$G^{\mathrm{RE}_a\mathrm{Ni}_b}_m = \frac{a}{a+b}\,{}^0G^{\chi}_{\mathrm{RE}} + \frac{b}{a+b}\,{}^0G^{\mathrm{fcc}}_{\mathrm{Ni}} + C + \mathrm{DT} \tag{6}$$

in which a and b represent the values of the stoichiometric numbers in these intermetallic compounds $\mathrm{RE}_a\mathrm{Ni}_b$. The parameters C and D are to be optimized in this work. χ refers to the standard reference phase of RE stable at 1 bar and 298.15 K.

In the Ce-Ni binary system, the molar Gibbs energy of Laves phase $CeNi_2$ is described using the two-sublattice model $(\mathrm{Ce, Ni})_{1/3}(\mathrm{Ce, Ni})_{2/3}$ according to the suggestion of Ansara et al. [43]. Based on the homogeneity range determined experimentally by Kim et al. [44], $CeNi_5$ phase is described using the two-sublattice model $(\mathrm{Ce, Ni})_{1/6}(\mathrm{Ce, Ni})_{5/6}$. The molar Gibbs energies of $CeNi_2$ and $CeNi_5$ can be represented by:

$$\begin{aligned} G^{\mathrm{CeNi_2}}_m =& y'_{\mathrm{Ce}}y''_{\mathrm{Ce}}\,{}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ce}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ni}}\,{}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ni}} \\ &+ y'_{\mathrm{Ce}}y''_{\mathrm{Ni}}\,{}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ni}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}\,{}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ce}} \\ &+ 0.3333RT(y'_{\mathrm{Ce}}\ln y'_{\mathrm{Ce}} + y'_{\mathrm{Ni}}\ln y'_{\mathrm{Ni}}) \\ &+ 0.6667RT(y''_{\mathrm{Ce}}\ln y''_{\mathrm{Ce}} + y''_{\mathrm{Ni}}\ln y''_{\mathrm{Ni}}) \\ &+ y'_{\mathrm{Ce}}y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}L^0_{\mathrm{Ce,Ni:Ce}} + y'_{\mathrm{Ce}}y'_{\mathrm{Ni}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ce,Ni:Ni}} \\ &+ y'_{\mathrm{Ce}}y''_{\mathrm{Ce}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ce:Ce,Ni}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ni:Ce,Ni}} \end{aligned} \tag{7}$$

$$\begin{aligned} G^{\mathrm{CeNi_5}}_m =& y'_{\mathrm{Ce}}y''_{\mathrm{Ce}}\,{}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ni}}\,{}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ni}} \\ &+ y'_{\mathrm{Ce}}y''_{\mathrm{Ni}}\,{}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ni}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}\,{}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce}} \\ &+ 0.1667RT(y'_{\mathrm{Ce}}\ln y'_{\mathrm{Ce}} + y'_{\mathrm{Ni}}\ln y'_{\mathrm{Ni}}) \\ &+ 0.8333RT(y''_{\mathrm{Ce}}\ln y''_{\mathrm{Ce}} + y''_{\mathrm{Ni}}\ln y''_{\mathrm{Ni}}) \\ &+ y'_{\mathrm{Ce}}y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}L^0_{\mathrm{Ce,Ni:Ce}} + y'_{\mathrm{Ce}}y'_{\mathrm{Ni}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ce,Ni:Ni}} \\ &+ y'_{\mathrm{Ce}}y''_{\mathrm{Ce}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ce:Ce,Ni}} + y'_{\mathrm{Ni}}y''_{\mathrm{Ce}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ni:Ce,Ni}} \end{aligned} \tag{8}$$

in which y'_{Ce}, y''_{Ce}, y'_{Ni} and y''_{Ni} are the site fractions of Ce and Ni in the first and second sublattice, respectively. ${}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ni}}$, ${}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ce}}$, ${}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ni}}$, ${}^0G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ce}}$, ${}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce}}$, ${}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ni}}$, ${}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ni}}$ and ${}^0G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce}}$ represent the Gibbs energy of possible end members for $CeNi_2$ and $CeNi_5$, while the parameters, $L^0_{\mathrm{Ce,Ni:Ce}}$, $L^0_{\mathrm{Ce,Ni:Ni}}$, $L^0_{\mathrm{Ce:Ce,Ni}}$ and $L^0_{\mathrm{Ni:Ce,Ni}}$ represent the interactions within each sublattice.

In the Ho-Ni binary system, the molar Gibbs energy of $HoNi_2$ is described using the two-sublattice model $(\mathrm{Ho})_{1/3}(\mathrm{Ho, Ni})_{2/3}$ considering its homogeneity range determined experimentally by Zhou et al. [45], and is expressed by:

$$\begin{aligned} G^{HoNi_2}_m =& y''_{\mathrm{Ho}}\,{}^0G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ho}} + y''_{\mathrm{Ni}}\,{}^0G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ni}} \\ &+ 0.6667RT(y''_{\mathrm{Ho}}\ln y''_{\mathrm{Ho}} + y''_{\mathrm{Ni}}\ln y''_{\mathrm{Ni}}) \\ &+ y''_{\mathrm{Ho}}y''_{\mathrm{Ni}}L^0_{\mathrm{Ho:Ho,Ni}} \end{aligned} \tag{9}$$

in which y''_{Ho} and y''_{Ni} are the site fractions of Ho and Ni in the second sublattice, respectively. ${}^0G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ho}}$ and ${}^0G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ni}}$ are the Gibbs energy of the end members for $HoNi_2$. The parameter $L^0_{\mathrm{Ho:Ho,Ni}}$ is the interaction parameter dependent on temperature.

Thermodynamic calculation

Based on the selected and reliable experimental data on phase equilibria and thermodynamic properties

(e.g., enthalpy of mixing of liquid alloys and enthalpy of formation of intermetallic compounds), the model parameters formulating the Gibbs energy of the phases in the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems were optimized using the PARROT module in Thermo-Calc® software [46]. Generally speaking, the experimental data on thermodynamic properties of the RE-Ni binary systems (e.g., Tb-Ni and Ho-Ni binary systems) are very limited due to various difficulties in the experiments (e.g., rare earth oxidation). Fortunately, the systematic trends found for the variation of thermodynamic properties of the RE-Mn and RE-Fe binary systems upon the change of the atomic number of rare earth metals [47–50] could be borrowed for the same properties in the RE-Ni systems because Ni as a 3d transition metal is similar to Mn and Fe. These trends were taken into account in assessing the RE-Ni binary systems in this work. Finally, the thermodynamic parameters of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems were obtained as shown in Tables 1, 2, 3, 4 and 5.

In addition, the phase equilibria and thermodynamic properties of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er, and Y) binary systems were calculated using the reported thermodynamic parameters in the literature in order to discuss systematically the variation trends of the phase equilibria and thermodynamic properties in the RE-Ni binary systems. The enthalpies of formation of the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) intermetallic compounds were summarized and given in the supplementary materials together with the crystal structure information. The comparison of the calculated results with the experimental data is shown in the following section.

Calculated results

La-Ni binary system

Figure 1 is the calculated La-Ni binary phase diagram compared with the experimental data [28, 29, 51–53] and the previous assessment results [29–32]. In this work, nine stable intermetallic compounds La_3Ni, La_7Ni_3, LaNi, La_2Ni_3, La_7Ni_{16}, $LaNi_3$, La_2Ni_7, La_5Ni_{19} and $LaNi_5$ were taken into account according to the experimental results determined by Yamamoto et al. [17], Dischinger and Schaller [31], Zhang et al. [52], Buschow and Van Mal [53], Vogel and Fülling [54] and Ivanchenko et al. [55]. The intermetallic compound La_4Ni_{17} reported by Dischinger and Schaller [31] has never been confirmed by others and thus was not considered in the present

Table 1 Thermodynamic parameters of the La-Ni binary system

Phase	Thermodynamic parameters	Reference
Liquid	${}^0L^{Liq}_{La,Ni} = -113820 + 9.6T$	This work
	${}^1L^{Liq}_{La,Ni} = 56900 - 25.1T$	
	${}^2L^{Liq}_{La,Ni} = -9500 + 12.5T$	
bcc(La)	${}^0L^{bcc}_{La,Ni} = 80000$	This work
fcc(La, Ni)	${}^0L^{fcc}_{La,Ni} = 80000$	This work
dhcp(La)	${}^0G^{dhcp}_{Ni} = {}^0G^{fcc}_{Ni} + 5000$	[30]
	${}^0L^{dhcp}_{La,Ni} = 80000$	This work
La_3Ni	$G^{La_3Ni}_m = -16548 + 0.731T + 0.75{}^0G^{dhcp}_{La} + 0.25{}^0G^{fcc}_{Ni}$	This work
La_7Ni_3	$G^{La_7Ni_3}_m = -19696 + 1.132 + 0.7{}^0G^{dhcp}_{La} + 0.3{}^0G^{fcc}_{Ni}$	This work
LaNi	$G^{LaNi}_m = -25998 - 1.4407T + 0.5{}^0G^{dhcp}_{La} + 0.5{}^0G^{fcc}_{Ni}$	This work
La_2Ni_3	$G^{La_2Ni_3}_m = -27341 + 0.362T + 0.4{}^0G^{dhcp}_{La} + 0.6{}^0G^{fcc}_{Ni}$	This work
La_7Ni_{16}	$G^{La_7Ni_{16}}_m = -28702 + 2.4621T + 0.3043{}^0G^{dhcp}_{La} + 0.6957{}^0G^{fcc}_{Ni}$	This work
$LaNi_3$	$G^{LaNi_3}_m = -28775 + 3.0547T + 0.25{}^0G^{dhcp}_{La} + 0.75{}^0G^{fcc}_{Ni}$	This work
La_2Ni_7	$G^{La_2Ni_7}_m = -28412 + 3.1854T + 0.2222{}^0G^{dhcp}_{La} + 0.7778{}^0G^{fcc}_{Ni}$	This work
La_5Ni_{19}	$G^{La_5Ni_{19}}_m = -28293 + 3.4733T + 0.2083{}^0G^{dhcp}_{La} + 0.7917{}^0G^{fcc}_{Ni}$	This work
$LaNi_5$	$G^{LaNi_5}_m = -27960 + 4.459T + 0.1667{}^0G^{dhcp}_{La} + 0.8333{}^0G^{fcc}_{Ni}$	This work

Table 2 Thermodynamic parameters of the Ce-Ni binary system

Phase	Thermodynamic parameters	Reference
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ce,Ni}} = -134053.248 + 9.22T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ce,Ni}} = 60037.3714 - 10.49T$	
bcc(Ce)	${}^{0}L^{bcc}_{Ce,Ni} = -56000$	This work
fcc(Ce, Ni)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}}/{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$ cited from SGTE database	[41]
Ce_7Ni_3	$G^{\mathrm{Ce_7Ni_3}}_{\mathrm{m}} = -24355 + 2.923T + 0.7{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
CeNi	$G^{\mathrm{CeNi}}_{\mathrm{m}} = -31185 - 1.571T + 0.5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$CeNi_2$	$G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ni}} = -37380 + 3.5976276T + 0.3333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.6667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
	$G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ce}} = 47380 - 3.5976276T + 0.6667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.3333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
	$G^{\mathrm{CeNi_2}}_{\mathrm{Ce:Ce}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 5000$	[68]
	$G^{\mathrm{CeNi_2}}_{\mathrm{Ni:Ni}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} + 5000$	[68]
$CeNi_3$	$G^{\mathrm{CeNi_3}}_{\mathrm{m}} = -34778 + 2.442T + 0.25{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.75{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ce_2Ni_7	$G^{\mathrm{Ce_2Ni_7}}_{\mathrm{m}} = -34056 + 2.608T + 0.2222{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.7778{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ce_5Ni_{19}	$G^{\mathrm{Ce_5Ni_{19}}}_{\mathrm{m}} = -33750 + 2.896T + 0.2083{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.7917{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$CeNi_5$	$G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ni}} = -32910 + 4.0514T + 0.1667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.8333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
	$G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce}} = 42910 - 4.0514T + 0.8333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 0.1667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
	$G^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} + 5000$	[68]
	$G^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ni}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} + 5000$	[68]
	${}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce,Ni:Ce}} = {}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce,Ni:Ni}} = -1143.7$	[68]
	${}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ce:Ce,Ni}} = {}^{0}L^{\mathrm{CeNi_5}}_{\mathrm{Ni:Ce,Ni}} = -35496.0$	[68]

Table 3 Thermodynamic parameters of the Nd-Ni binary system

Phase	Thermodynamic parameters	Reference
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Nd,Ni}} = -131550 + 23.1T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Nd,Ni}} = 60300 - 22.8T$	
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Nd,Ni}} = -4100 + 8.6\mathrm{T}$	
bcc(Nd)	${}^{0}L^{\mathrm{bcc}}_{\mathrm{Nd,Ni}} = 10000$	This work
dhcp(Nd)	${}^{0}G^{\mathrm{dhcp}}_{\mathrm{Ni}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} + 5000$	This work
	${}^{0}L^{\mathrm{dhcp}}_{\mathrm{Nd,Ni}} = 10000$	
fcc(Ni)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Nd}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 5000$	This work
	${}^{0}L^{\mathrm{fcc}}_{\mathrm{Nd,Ni}} = 80000$	
Nd_3Ni	$G^{\mathrm{Nd_3Ni}}_{\mathrm{m}} = -22010 + 5.4235\mathrm{T} + 0.75{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.25{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Nd_7Ni_3	$G^{\mathrm{Nd_7Ni_3}}_{\mathrm{m}} = -26770 + 7.725\mathrm{T} + 0.7{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
NdNi	$G^{\mathrm{NdNi}}_{\mathrm{m}} = -29880 + 0.545\mathrm{T} + 0.5{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$NdNi_2$	$G^{\mathrm{NdNi_2}}_{\mathrm{m}} = -30681 + 1.65183713\mathrm{T} + 0.3333{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.6667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$NdNi_3$	$G^{\mathrm{NdNi_3}}_{\mathrm{m}} = -30211 + 3.06707294\mathrm{T} + 0.25{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.75{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Nd_2Ni_7	$G^{\mathrm{Nd_2Ni_7}}_{\mathrm{m}} = -29905 + 3.61348781\mathrm{T} + 0.2222{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.7778{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$NdNi_5$	$G^{\mathrm{NdNi_5}}_{\mathrm{m}} = -28500 + 4.636\mathrm{T} + 0.1667{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.8333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Nd_2Ni_{17}	$G^{\mathrm{Nd_2Ni_{17}}}_{\mathrm{m}} = -12252 - 0.855T + 0.1053{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 0.8947{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work

Table 4 Thermodynamic parameters of the Tb-Ni binary system

Phase	Thermodynamic parameters	Reference
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ni,Tb}} = -165000 + 29.5T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ni,Tb}} = -71000 + 27.8T$	
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ni,Tb}} = 18000$	
bcc(Tb)	${}^{0}G^{\mathrm{bcc}}_{\mathrm{Tb}}$ cited from SGTE database	[41]
hcp(Tb)	${}^{0}G^{\mathrm{bcc}}_{\mathrm{Tb}}$ cited from SGTE database	[41]
fcc(Ni)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$ cited from SGTE database	[41]
$NiTb_3$	$G_m^{\mathrm{NiTb_3}} = -18509 - 1.593\mathrm{T} + 0.25{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.75{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
Ni_2Tb_3	$G_m^{\mathrm{Ni_2Tb_3}} = -30050 - 0.681\mathrm{T} + 0.4{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.6{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
NiTb	$G_m^{\mathrm{NiTb}} = -38200 + 2.185\mathrm{T} + 0.5{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
Ni_2Tb	$G_m^{\mathrm{Ni_2Tb}} = -40000 + 4.143\mathrm{T} + 0.6667{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.3333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
Ni_3Tb	$G_m^{\mathrm{Ni_3Tb}} = -39933.3805 + 6.66452548\mathrm{T} + 0.75{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.25{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
Ni_7Tb_2	$G_m^{\mathrm{Ni_7Tb_2}} = -38463.7678 + 6.800\mathrm{T} + 0.7778{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.2222{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
Ni_5Tb	$G_m^{\mathrm{Ni_7Tb_2}} = -32086 + 5.45123969\mathrm{T} + 0.8333{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.1667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work
$Ni_{17}Tb_2$	$G_m^{\mathrm{Ni_{17}Tb_2}} = -21418.9812 + 4.08760457\mathrm{T} + 0.8947{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ni}} + 0.1053{}^{0}G^{\mathrm{fcc}}_{\mathrm{Tb}}$	This work

Table 5 Thermodynamic parameters of the Ho-Ni binary system

Phase	Thermodynamic parameters	Reference
Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ho,Ni}} = -170100 + 26.6T$	This work
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ho,Ni}} = 35000 - 13.8T$	
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ho,Ni}} = 16000$	
bcc(Ho)	${}^{0}G^{\mathrm{bcc}}_{\mathrm{Ho}}$ cited from SGTE database	[41]
hcp(Ho)	${}^{0}G^{\mathrm{bcc}}_{\mathrm{Ho}}$ cited from SGTE database	[41]
fcc(Ni)	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$ cited from SGTE database	[41]
Ho_3Ni	$G_m^{\mathrm{Ho_3Ni}} = -23534 + 1.644\mathrm{T} + 0.75{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.25{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ho_3Ni_2_LT	$G_m^{\mathrm{Ho_3Ni_2_LT}} = -38341 + 5.516\mathrm{T} + 0.6{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.4{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ho_3Ni_2_HT	$G_m^{\mathrm{Ho_3Ni_2_HT}} = -28176 - 4.7106\mathrm{T} + 0.6{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.4{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
HoNi	$G_m^{\mathrm{HoNi}} = -48386 + 8.1389\mathrm{T} + 0.5{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$HoNi_2$	$G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ni}} = -43802 + 6.00388433\mathrm{T} + 0.3333{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.6667{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
	$G^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ho}} = 10000 + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} \quad {}^{0}L^{\mathrm{HoNi_2}}_{\mathrm{Ho:Ho,Ni}} = -32000 + 8T$	[40]
$HoNi_3$	$G_m^{\mathrm{HoNi_3}} = -41377 + 7.0721699\mathrm{T} + 0.25{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.75{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ho_2Ni_7	$G_m^{\mathrm{Ho_2Ni_7}} = -40596 + 7.82821171\mathrm{T} + 0.2222{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.7778{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
$HoNi_5$	$G_m^{\mathrm{HoNi_5}} = -34500 + 7.034\mathrm{T} + 0.1667{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.8333{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work
Ho_2Ni_{17}	$G_m^{\mathrm{Ho_2Ni_{17}}} = -22230 + 4.663T + 0.1053{}^{0}G^{\mathrm{hcp}}_{\mathrm{Ho}} + 0.8947{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}}$	This work

calculation. Although La_5Ni_{19} was reported to be unstable at low temperatures in the literature [17, 28, 56], it was assumed to be stable at and above room temperature in this work because the decomposition temperature of La_5Ni_{19} was not determined [28]. In addition, $LaNi_5$ is regarded as a stoichiometric compound in this work because the homogeneity range of $LaNi_5$ is less than 1.5 at.% according to the experimental results determined by Buschow and Van Mal [53] and Fartushna et al. [28]. Compared with the calculated results by An et al. [29], Du et al. [30], Dischinger and Schaller [31], and Liu et al. [32] in Fig. 1a, the calculated La-Ni binary phase diagram in this work is in much better

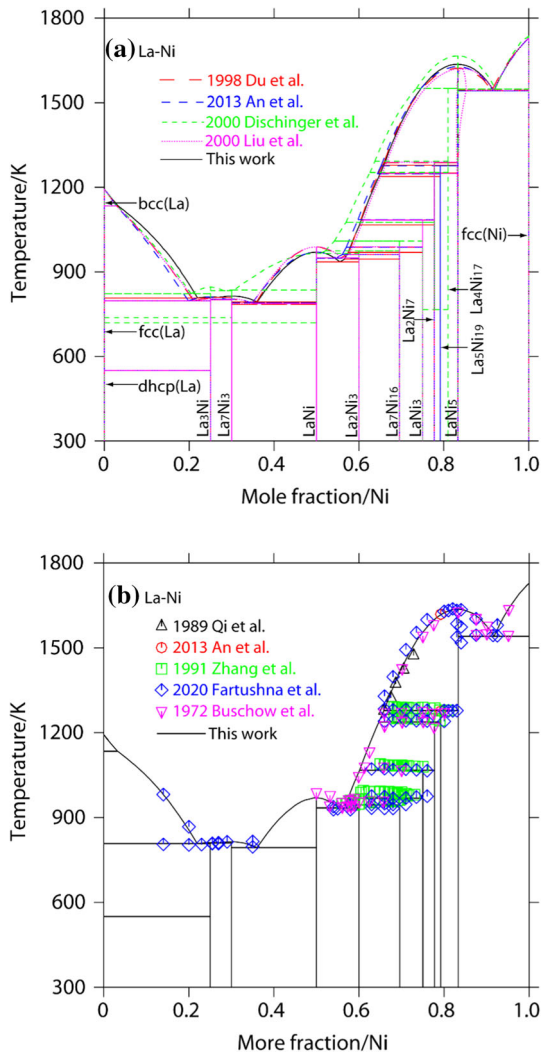


Figure 1 The calculated La-Ni binary phase diagram in comparison with the experimental data [28, 29, 51–53] and the prior calculation results [29–32].

agreement with the experimental data [28, 29, 51–53] as shown in Fig. 1b. In particular, the calculated liquidus are consistent well with the experimental results [28, 29, 51, 53]. Table 6 shows the calculated temperatures of invariant reactions in the La-Ni binary system in this work with the experimental data [28, 31, 52, 53] and the calculated results [29–31]. It is obvious that the calculated temperatures of five eutectic reactions ($L \leftrightarrow LaNi_5$ + fcc(Ni), $L \leftrightarrow LaNi + La_2Ni_3$, $L \leftrightarrow LaNi + La_7Ni_3$, $L \leftrightarrow La_3Ni + La_7Ni_3$ and $L \leftrightarrow$ fcc(La) + La_3Ni) in the literature [29–32] are different. Based on the difference between the experimental data [29, 51, 52] and the calculated results [30–32], Fartushna et al. [28] re-investigated experimentally the La-Ni binary phase diagram in the whole composition range. During the present calculation, the temperatures of five eutectic reactions in the La-Ni binary system were optimized mainly according to the experimental data measured by Fartushna et al. [28]. As can be seen in Table 6, the calculated results in this work are in good agreement with the experimental data [28].

Figure 2 shows the calculated enthalpy of mixing of the liquid La-Ni alloys at 1800 K with the experimental data [57, 58]. Figure 3 is the calculated partial enthalpy of mixing of the liquid La-Ni alloys at 1800 K with the experimental data [58] and the earlier calculation results [29, 30]. The enthalpies of mixing of the liquid La-Ni alloys at 1376 K were determined by Watanabe and Kleppa [57], while the enthalpies of mixing and partial enthalpies of mixing of the liquid La-Ni alloys at 1430 K and 1800 K were measured by Subotenko et al. [58] using isoperibol calorimetry. The calculated enthalpy of mixing and partial enthalpy of mixing of the liquid La-Ni alloys in this work are in better agreement with the experimental data [57, 58].

Figure 4 is the calculated enthalpy of formation of the La-Ni intermetallic compounds at 298 K in comparison with the experimental data [57, 59–64] and the calculation results [29, 30, 65, 66]. It can be seen that our calculation results are in good agreement with the experimental data [57, 59–64], while the calculation results from Dischinger and Schaller [31], Open Quantum Materials Database [65] and Materials Project [66] are much more negative. Finally, the calculated heat content for $LaNi_5$ is depicted in Fig. 5 together with the experimental data measured by Yamaguchi et al. [67] using a drop calorimeter. The calculated results are consistent well with the experimental data [67].

Ce-Ni binary system

Figure 6 is the calculated Ce-Ni binary phase diagram in comparison with the experimental data [44, 51, 54, 68–73] and the previous calculation results [68, 74, 75]. It is obvious that our results are quite close to the most recent assessment results by Xiong et al. [68] and agree reasonably well with the experimental data [44, 51, 54, 68–73]. The calculated invariant reactions in the Ce-Ni binary system are shown in Table 7. In this work, the new intermetallic compound Ce_5Ni_{19} found by Mardani et al. [33] was considered and it is calculated to be formed by the

Table 6 Invariant reactions in the La-Ni binary system

Reactions	Reaction type	Temperature (K)	Composition(x_{Ni}^{L})	Reference
L ↔ $LaNi_5$ + fcc(Ni)	Eutectic	1541	–	[29] (cal.)
		1543	–	[53] (exp.)
		1543	0.913	[30] (cal.)
		1548	–	[31] (cal.)
		1552	–	[31] (exp.)
		1544	0.91	[28] (exp.)
		1542	0.917	[32] (cal.)
		1541.2	0.918	This work (cal.)
L ↔ $LaNi_5$	Congruent	1623	0.833	[53] (exp.)
		1622	0.833	[30] (cal.)
		1666	0.833	[31] (cal.)
		1627	0.833	[29] (cal.)
		1663	0.833	[31] (exp.)
		1634	0.833	[28] (exp.)
		1621	0.833	[32] (cal.)
		1635.9	0.833	This work (cal.)
L + $LaNi_5$ ↔ La_5Ni_{19}	Peritectic	1276	–	[29] (cal.)
		1276	–	[29] (exp.)
		~ 1273	–	[17] (exp.)
		1280	–	[28] (exp.)
		1277.7	0.659	This work (cal.)
L + La_5Ni_{19} ↔ La_2Ni_7	Peritectic	1247	–	[29] (cal.)
		1241	–	[28] (exp.)
		1238.4	0.647	This work (cal.)
L + La_2Ni_7 ↔ $LaNi_3$	Peritectic	1084	–	[29] (cal.)
		1084	–	[52] (exp.)
		1228	–	[53] (exp.)
		1084	0.605	[30] (cal.)
		1075	–	[31] (cal.)
		1086	–	[31] (exp.)
		1069	–	[28] (exp.)
		1084	0.611	[32] (cal.)
		1066.2	0.601	This work (cal.)
L + $LaNi_3$ ↔ La_7Ni_{16}	Peritectic	987	–	[29] (cal.)
		987	–	[52] (exp.)
		1068	–	[53] (exp.)
		987	0.581	[30] (cal.)
		1010	–	[31] (cal.)
		998	–	[31] (exp.)
		971	–	[28] (exp.)
		987	0.586	[32] (cal.)
		968.2	0.575	This work (cal.)
L + La_7Ni_{16} ↔ La_2Ni_3	Peritectic	961	–	[29] (cal.)
		961	–	[52] (exp.)
		963	–	[53] (exp.)
		960	0.572	[30] (cal.)
		974	–	[31] (cal.)
		974	–	[31] (exp.)
		947	–	[28] (exp.)
		960	0.575	[32] (cal.)
		944.1	0.566	This work (cal.)

Table 6 continued

Reactions	Reaction type	Temperature (K)	Composition(x^L_{Ni})	Reference
L ↔ LaNi + La_2Ni_3	Eutectic	948	–	[29] (cal.)
		948	–	[52] (exp.)
		943	–	[53] (exp.)
		949	0.557	[30] (cal.)
		964	–	[31] (cal.)
		963	–	[31] (exp.)
		931	0.544	[28] (exp.)
		949	0.560	[32] (cal.)
		934.2	0.553	This work (cal.)
L ↔ LaNi	Congruent	988	0.500	[52] (exp.)
		988	0.500	[53] (exp.)
		965	0.500	[31] (cal.)
		992	0.500	[31] (exp.)
		969	0.500	[29] (cal.)
		988	0.500	[32] (cal.)
		969.6	0.500	This work (cal.)
L ↔ LaNi + La_7Ni_3	Eutectic	790	–	[29] (cal.)
		784	0.355	[30] (cal.)
		836	–	[31] (cal.)
		833	–	[31] (exp.)
		796	0.344	[28] (exp.)
		788	0.349	[32] (cal.)
		793.6	0.360	This work (cal.)
L ↔ La_7Ni_3	Congruent	803	0.300	[30] (cal.)
		837	0.300	[31] (cal.)
		803	0.300	[29] (cal.)
		848	0.300	[31] (exp.)
		814	0.300	[28] (exp.)
		805	0.300	[32] (cal.)
		814.0	0.300	This work (cal.)
L ↔ La_3Ni + La_7Ni_3	Eutectic	803	–	[29] (cal.)
		801	0.279	[30] (cal.)
		834	–	[31] (cal.)
		818	–	[31] (exp.)
		807	0.254	[28] (exp.)
		801	0.276	[32] (cal.)
		809.6	0.269	This work (cal.)
L ↔ La_3Ni	Congruent	805	0.25	[30] (cal.)
		846	0.25	[31] (cal.)
		810	0.25	[29] (cal.)
		843	0.25	[31] (exp.)
		805	0.25	[32] (cal.)
		810.9	0.25	This work (cal.)
L ↔ fcc(La) + La_3Ni	Eutectic	798	–	[29] (cal.)
		797	0.211	[30] (cal.)
		822	–	[31] (cal.)
		834	–	[31] (exp.)
		804	0.23	[28] (exp.)
		798	0.220	[32] (cal.)
		807.1	0.219	This work (cal.)

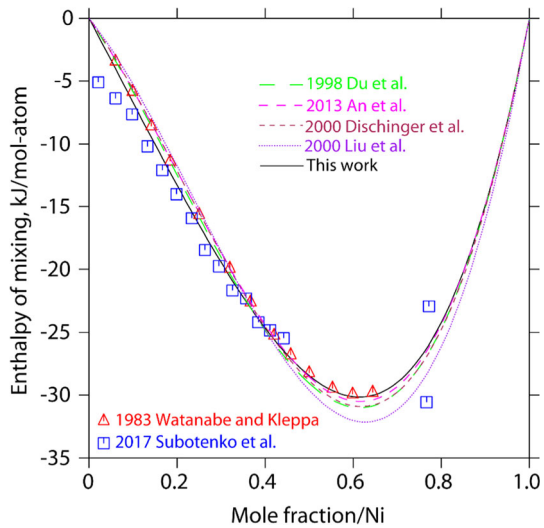


Figure 2 The calculated enthalpy of mixing of the liquid La-Ni alloy at 1800 K in comparison with the experimental data [57, 58] and the prior calculation results [29–32]. (Reference states: liquid La and liquid Ni).

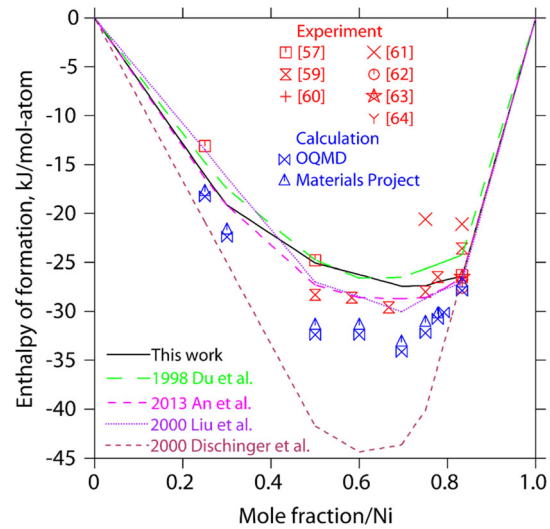


Figure 4 The calculated enthalpy of formation of the La-Ni intermetallic compounds at 298 K in comparison with the experimental data [57, 59–64] and some available calculation results [29–32, 65, 66]. (Reference states: dhcp-La and fcc-Ni).

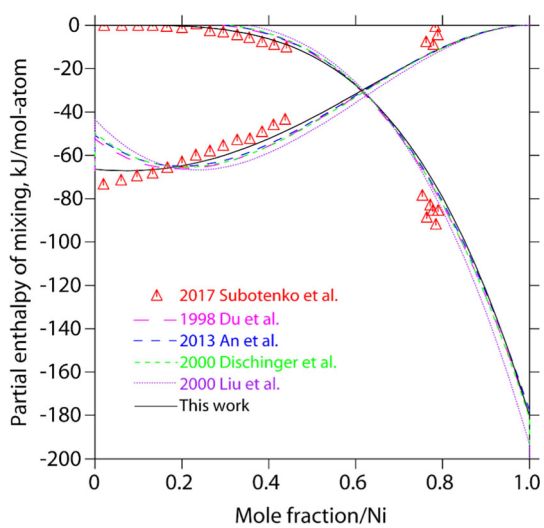


Figure 3 The calculated partial enthalpy of mixing of the liquid La-Ni alloy at 1800 K in comparison with the experimental data [58] and the prior calculation results [29–32]. (Reference states: liquid La and liquid Ni).

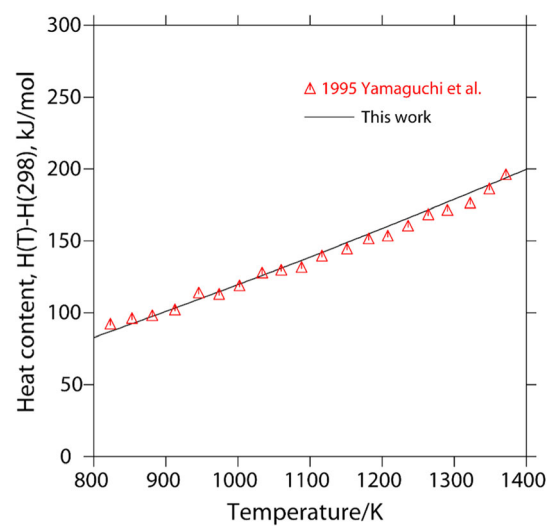


Figure 5 The calculated heat content of $LaNi_5$ in comparison with the experimental data [67].

peritectic reaction (L + $CeNi_5$ ↔ Ce_5Ni_{19}) at 1396 K. Furthermore, $CeNi_5$ has a certain homogeneity range determined by Kim et al. [44], which was also considered in this work. In addition, seven stable intermetallic compounds Ce_7Ni_3, CeNi, $CeNi_2$, $CeNi_3$, Ce_2Ni_7, Ce_5Ni_{19} and $CeNi_5$ were included in the present calculation according to the experimental results reported by Kim et al. [44], Qi et al. [51], Xiong et al. [68], Pish et al. [69], Gebhart et al. [70], Vogel et al. [54], Duisemaliev et al. [72] and Perkins et al. [73].

The enthalpy of mixing of the liquid Ce-Ni alloys was determined experimentally by Sudavtsov et al. [76], Nikolaenko and Vlasova [77], and recently by Ivanov et al. [78]. Based on the experimental data [77, 78], the enthalpy of mixing of the liquid Ce-Ni

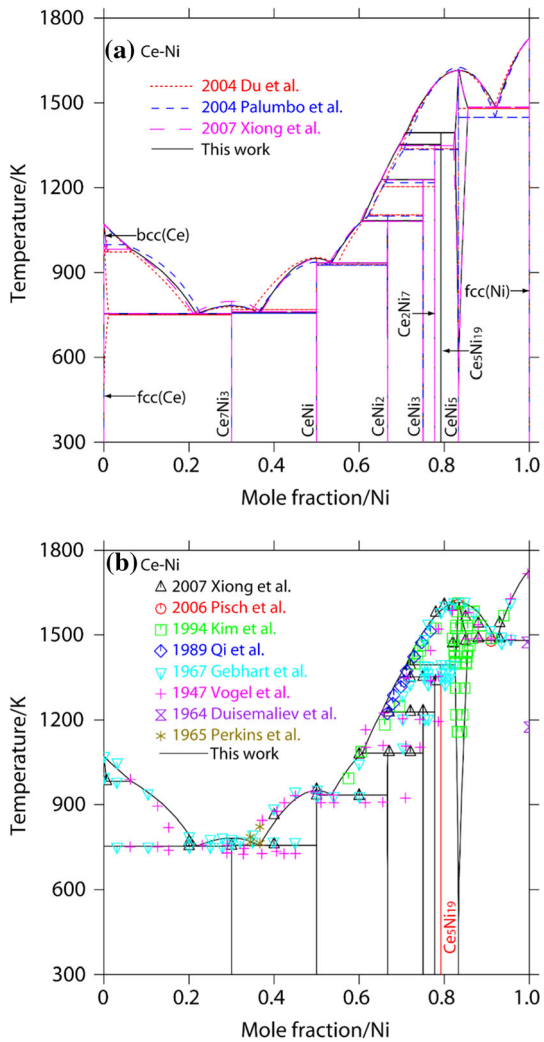


Figure 6 The calculated Ce-Ni binary phase diagram in comparison with the experimental data [44, 51, 54, 68–70, 72, 73] and the prior calculation results [68, 74, 75].

alloys was assessed in this work. Figures 7 and 8 are the calculated enthalpy of mixing and partial enthalpy of mixing for the liquid Ce-Ni alloys at 1820 K compared with the experimental data [77, 78] and the assessed results [68, 74, 75], respectively. It can be seen that the assessed results by Du et al. [74] are much more positive, while those by Palumbo et al. [75] are more negative. The calculated results in this work are very close to the calculated results by Xiong et al. [68].

Figure 9 shows the calculated enthalpy of formation of the Ce-Ni intermetallic compounds at 298 K together with the experimental data [67, 75, 79–83] and the calculated results [65, 66, 68, 74, 75]. The calculation results by Du et al. [74] are much more positive in the composition range of 50–83.3 at.% Ni, while those by Palumbo et al. [75] are much more negative. The calculated results in this work are similar to the calculated results by Xiong et al. [68]. It was found that the DFT calculation results from Open Quantum Materials Database [65] are quite different from the experimental data [67, 75, 79–83] and the assessment results in this work, while those from Materials Project [66] are mostly much closer. In addition, as shown in Fig. 10, the calculated heat content for $CeNi_5$ in this work is in good agreement with the experimental data measured by Yamaguchi et al. [67].

Nd-Ni binary system

Figure 11 shows the calculated Nd-Ni binary phase diagram in comparison with the experimental data [35, 51, 84] and the previous assessment results [36, 37]. It can be seen in Fig. 11a, b that the calculated results in this work are in much better agreement with the experimental data [35, 51, 84]. Furthermore, the calculated stable temperature range of Nd_2Ni_{17} in this work is consistent with the results by Pan and Zheng [84] and is different from the narrow temperature range suggested by Huang et al. [35]. Although the intermetallic compound Nd_5Ni_{19} was found by Lemort et al. [85], the type and temperature of the reaction for the formation of Nd_5Ni_{19} in the La-Ni binary system were not determined. Therefore, Nd_5Ni_{19} was not taken into account in the present calculation. Eight stable intermetallic compounds, Nd_3Ni, Nd_7Ni_3, NdNi, $NdNi_2$, $NdNi_3$, Nd_2Ni_7, $NdNi_5$ and Nd_2Ni_{17}, were taken into account in this work according to the experimental results determined by Huang et al. [35], Qi et al. [51], Pan and Zheng [84]. It should be mentioned that the polycrystalline transition of Nd_2Ni_7 determined experimentally by Huang et al. [35] was not confirmed in other work [34, 36, 51, 84] and thus was not considered in this work. Table 8 lists the calculated results of the invariant reactions in the Nd-Ni binary system together with the experimental data [35, 84] and the prior calculation results [34–38]. It can be seen that the calculated temperatures of the invariant reactions in this work are in much better agreement with the experimental data [35] and the calculated results [36] and show the obvious deviation from the experimental results [84] and the assessment results [37].

Table 7 Invariant reactions in the Ce–Ni binary system

Reactions	Reaction type	Temperature (K)	Composition (x_{Ni}^{L})	Reference
bcc(Ce) ↔ fcc(Ce) + L	Metatectic	983	0.0058	[68] (cal.)
		982.7	0.0057	This work (cal.)
L ↔ fcc(Ce) + Ce_7Ni_3	Eutectic	743	0.191	[54] (exp.)
		750	0.220	[70] (exp.)
		756	0.225	[75] (cal.)
		757	0.210	[75] (exp.)
		750	0.207	[74] (cal.)
		752	0.213	[69] (cal.)
		754	–	[68] (exp.)
		756	0.210	[68] (cal.)
		753.6	0.218	This work (cal.)
L ↔ Ce_7Ni_3	Congruent	758	0.300	[54] (exp.)
		778	0.300	[70] (exp.)
		784	0.300	[75] (cal.)
		778	0.300	[75] (cal.)
		778	0.300	[69] (cal.)
		781.8	0.300	This work (cal.)
L ↔ Ce_7Ni_3 + CeNi	Eutectic	771	0.330	[75] (exp.)
		723	0.328	[54] (exp.)
		768	0.330	[70] (exp.)
		763	0.344	[73] (exp.)
		756	0.360	[75] (cal.)
		768	0.348	[74] (cal.)
		773	0.334	[69] (cal.)
		760	–	[68] (exp.)
		760	0.369	[68] (cal.)
		757	–	[33] (exp.)
		756.5	0.365	This work (cal.)
L ↔ CeNi	Congruent	943	0.500	[54] (exp.)
		953	0.500	[70] (exp.)
		937	0.500	[75] (cal.)
		953	0.500	[74] (cal.)
		953	0.500	[69] (cal.)
		950.1	0.500	This work (cal.)
L ↔ CeNi + $CeNi_2$	Eutectic	928	–	[75] (exp.)
		908	0.529	[54] (exp.)
		928	0.550	[70] (exp.)
		926	0.530	[75] (cal.)
		928	0.548	[74] (cal.)
		930	0.547	[69] (cal.)
		931	–	[68] (exp.)
		931	0.531	[68] (cal.)
		933.6	0.535	This work (cal.)

Table 7 continued

Reactions	Reaction type	Temperature (K)	Composition (x^L_{Ni})	Reference
L + $CeNi_3$ ↔ $CeNi_2$	Peritectic	1103	–	[54] (exp.)
		1203	0.620	[70] (exp.)
		1098	0.620	[75] (cal.)
		1103	0.625	[74] (cal.)
		1103	0.641	[69] (cal.)
		1086	–	[68] (exp.)
		1084	0.603	[68] (cal.)
		1083	–	[33] (exp.)
		1083.0	0.608	This work (cal.)
L + Ce_2Ni_7 ↔ $CeNi_3$	Peritectic	1233	–	[54] (exp.)
		1203	0.690	[70] (exp.)
		1218	0.660	[75] (cal.)
		1203	0.655	[74] (cal.)
		1233	0.684	[69] (cal.)
		1228	–	[68] (exp.)
		1230	0.651	[68] (cal.)
		1230	–	[33] (exp.)
		1228.3	0.653	This work (cal.)
L + Ce_5Ni_{19} ↔ Ce_2Ni_7	Peritectic	1355	–	[33] (exp.)
		1352.9	0.694	This work (cal.)
L + $CeNi_5$ ↔ Ce_5Ni_{19}	Peritectic	1396	–	[33] (exp.)
		1394.1	0.710	This work (cal.)
L ↔ $CeNi_5$	Congruent	1593	0.833	[54] (exp.)
		1598	0.833	[70] (exp.)
		1625	0.833	[75] (cal.)
		1613	0.833	[74] (cal.)
		1610	0.833	[69] (cal.)
		1608	0.833	[69] (exp.)
		1614.7	0.833	This work (cal.)
L ↔ $CeNi_5$ + fcc(Ni)	Eutectic	1477	0.910	[69] (exp.)
		1481	0.917	[69] (cal.)
		1488	0.911	[44] (exp.)
		1483	0.911	[54] (exp.)
		1478	0.910	[70] (exp.)
		1449	0.920	[75] (cal.)
		1480	0917	[74] (cal.)
		1488	0.919	[68] (exp.)
		1484	0.922	[68] (cal.)
		1480.2	0.922	This work (cal.)

The reason for the deviation would be that the experimental results determined by Huang et al. [35] were given much higher weight in the present calculation considering the reliability of the experimental data.

The mixing enthalpy of the liquid Nd-Ni alloys in the composition range of 0–50 at.% Ni at 1733 K and 55–100 at.% Ni at. 1773 K was determined recently by Sudavtsova et al. [38] by means of isoperibolic calorimetry. Figures 12 and 13 show the calculated enthalpy of mixing and partial enthalpy of mixing of the liquid Nd-Ni alloys at 1750 K in comparison with the experimental data [38] and the earlier assessment results [36, 37], respectively. The calculated results in

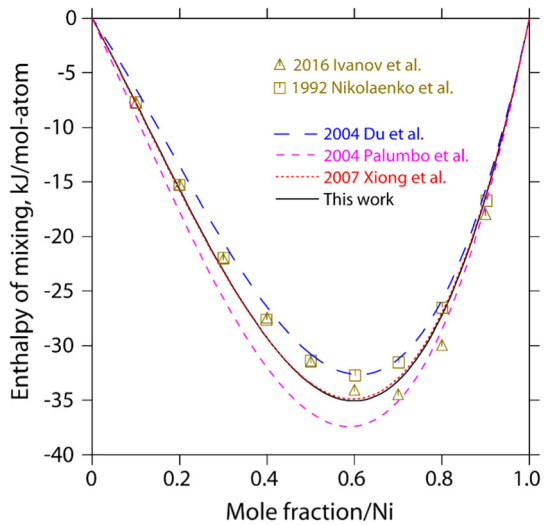


Figure 7 The calculated enthalpy of mixing of the liquid Ce-Ni alloys at 1820 K in comparison with the experimental data [77, 78] and the prior calculation results [68, 74, 75]. (Reference states: liquid Ce and liquid Ni).

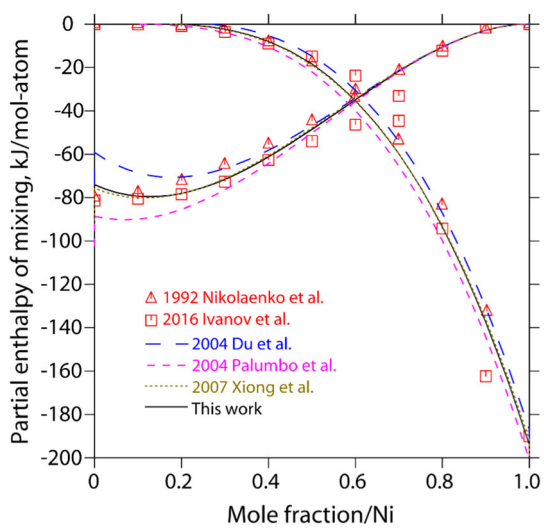


Figure 8 The calculated partial enthalpy of mixing of the liquid Ce-Ni alloys at 1820 K in comparison with the experimental data [77, 78] and the prior calculation results [68, 74, 75]. (Reference states: liquid Ce and liquid Ni).

this work are in much better agreement with the experimental data [38] than the prior assessment results [36, 37]. Although the calculation results by Hussain et al. [34] are in good agreement with the experimental results determined by Sudavtsova et al. [38], their calculated Nd-Ni binary phase diagram deviates obviously from the experimental data [35, 51, 84].

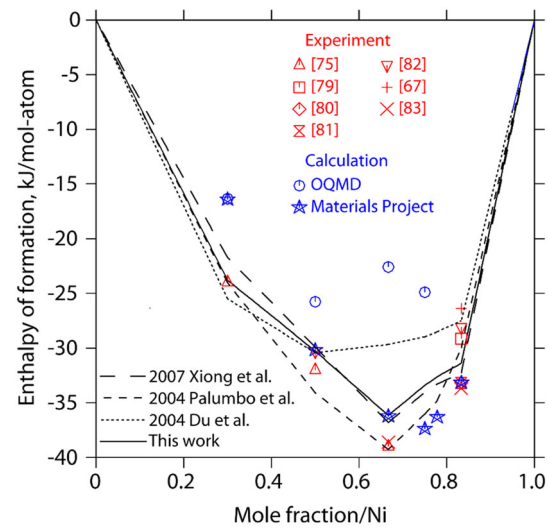


Figure 9 The calculated enthalpy of formation of the Ce-Ni intermetallic compounds at 298 K in comparison with the experimental data [67, 75, 79–83] and the prior calculation results [65, 66, 68, 74, 75]. (Reference states: fcc-Ce and fcc-Ni).

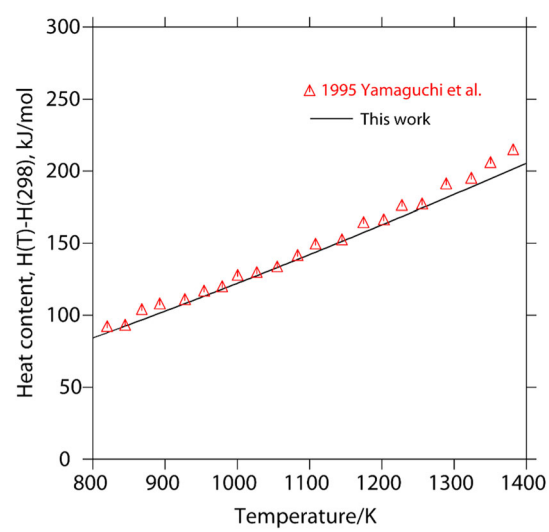


Figure 10 The calculated heat content of $CeNi_5$ in comparison with the experimental data [67].

As shown in Fig. 14, the calculated enthalpy of formation of the Nd-Ni intermetallic compounds in this work agrees reasonably well with the reported DFT data [65, 66] but not so much with the prior assessment results [36, 37] in the middle of the composition range. In Fig. 15, the calculated heat content for $NdNi_5$ in this work is shown in excellent agreement with the experimental data from Yamaguchi et al. [67].

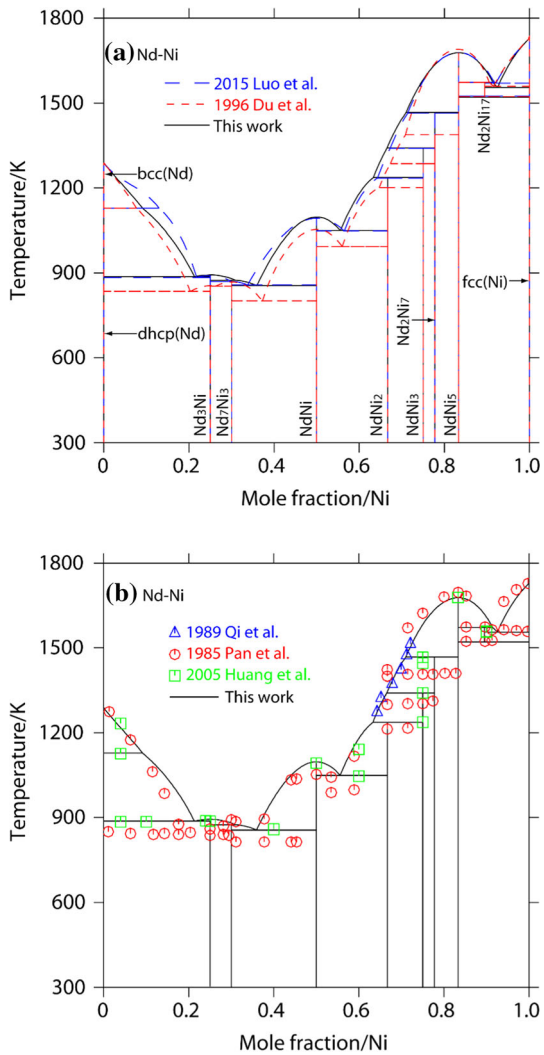


Figure 11 The calculated Nd-Ni binary phase diagram in comparison with the experimental data [35, 51, 84] and the prior calculation results [36, 37].

Tb-Ni binary system

Figure 16 is the calculated Tb-Ni binary phase diagram in comparison with the experimental data [39, 86] and the earlier assessment results [86]. Table 9 shows the calculated and measured temperatures of the invariant reactions in the Tb-Ni binary system. According to the experimental results determined by Rong and Schaller [86], and Yao et al. [39] using electromotive force measurements (EMF), differential thermal analysis (DTA) and X-ray diffraction (XRD), there are eight intermetallic compounds, Tb_2Ni_{17}, $TbNi_5$, Tb_2Ni_7, $TbNi_3$, $TbNi_2$, TbNi, Tb_3Ni_2 and Tb_3Ni in the Tb-Ni binary system, which were accepted in this work. The calculated Tb-Ni binary phase diagram in this work shows that $TbNi_2$ and Tb_2Ni_{17} are stable at low temperature in contrast to the calculated results by Rong and Schaller [86]. The temperature of the peritectic reaction (L + TbNi ↔ Tb_3Ni_2) determined by Yao et al. [39] is quite different from the experimental data measured by Rong and Schaller [86]. The calculated temperature of the peritectic reaction (L + TbNi ↔ Tb_3Ni_2) in this work is consistent with the measured data by Yao et al. [39].

Figure 17 shows the calculated enthalpy of mixing of the liquid Tb-Ni alloys at 1820 K in comparison with the earlier calculation results of Rong and Schaller [86]. The calculated results in this work are less negative than the earlier work [86]. Figure 18 is the calculated enthalpy of formation of the Tb-Ni intermetallic compounds at 298 K in comparison with the experimental data [82, 87] and the available calculation results [65, 66, 86]. Our calculated results are in good agreement with the experimental data [82, 87] and the DFT data from Materials Project [66]. It was noted that the absolute values for the enthalpy of formation of Tb_2Ni_7 and $TbNi_2$ from Open Quantum Materials Database [65] are much smaller than our results and those from Materials Project [66].

Ho-Ni binary system

Figure 19 is the calculated Ho-Ni binary phase diagram in comparison with the experimental data [45] and the prior calculation results [40]. The calculated and experimental invariant reactions are shown in Table 10. Based on the experimental results determined by Zhou et al. [45] using X-ray diffraction, differential thermal analysis, optical microscopy and electron probe microanalysis techniques, there are eight intermetallic compounds in the Ho-Ni binary system: Ho_3Ni, Ho_3Ni_2 (Ho_3Ni_2_LT at low temperatures and Ho_3Ni_2_HT at high temperatures), HoNi, $HoNi_2$, $HoNi_3$, Ho_2Ni_7, $HoNi_5$, and Ho_2Ni_{17}, and all of them were accepted in this work. As shown in Fig. 19b, the calculated Ho-Ni binary phase diagram agrees well reasonably with the experimental results [45]. It was noted that the calculated temperature of the invariant reaction L ↔ HoNi + $HoNi_2$ is much higher than the experimental data [45] and slightly lower than the previous calculation results [40]. This

Table 8 Invariant reactions in the Nd–Ni binary system

Reactions	Reaction type	Temperature (K)	Composition (x^L_{Ni})	Reference
L + bcc(Nd) ↔ dhcp(Nd)	Peritectic	1126	0.130	[35] (cal.)
		1126	–	[35] (exp.)
		1136	–	[84] (exp.)
		1128	0.077	[37] (cal.)
		1128	0.130	[36] (cal.)
		1135	0.072	[38] (cal.)
		1128.0	0.091	This work (cal.)
L ↔ Nd_3Ni + dhcp(Nd)	Eutectic	884	0.218	[35] (cal.)
		884	–	[35] (exp.)
		874.1	0.193	[34] (cal.)
		843	~ 0.190	[84] (exp.)
		834	0.203	[37] (cal.)
		884	0.218	[36] (cal.)
		883	0.218	[38] (cal.)
		887.0	0.213	This work (cal.)
L ↔ Nd_3Ni	Congruent	891	0.250	[35] (exp.)
		891	0.250	[35] (cal.)
		891.4	0.250	[34] (cal.)
		863	0.250	[84] (exp.)
		855	0.250	[37] (cal.)
		891	0.250	[36] (cal.)
		890	0.250	[38] (cal.)
		892.9	0.250	This work (cal.)
L + Nd_3Ni ↔ Nd_7Ni_3	Peritectic	871	–	[35] (exp.)
		869	0.301	[35] (cal.)
		868.6	–	[34] (cal.)
		838	–	[84] (exp.)
		853	–	[37] (cal.)
		869	0.301	[36] (cal.)
		869	0.303	[38] (cal.)
		873.8	0.310	This work (cal.)
L ↔ Nd_7Ni_3 + NdNi	Eutectic	858	–	[35] (exp.)
		857	0.338	[35] (cal.)
		867.3	0.321	[34] (cal.)
		813	~ 0.350	[84] (exp.)
		801	0.372	[37] (cal.)
		857	0.338	[36] (cal.)
		857	0.338	[38] (cal.)
		855.3	0.359	This work (cal.)
L ↔ NdNi	Congruent	1092	0.500	[35] (cal.)
		1092	0.500	[35] (exp.)
		1090.6	0.500	[34] (cal.)
		1053	0.500	[84] (exp.)
		1053	0.500	[37] (cal.)
		1092	0.500	[36] (cal.)
		1096	0.500	[38] (cal.)
		1096.8	0.500	This work (cal.)

Table 8 continued

Reactions	Reaction type	Temperature (K)	Composition (x^L_{Ni})	Reference
L ↔ NdNi + $NdNi_2$	Eutectic	1046	–	[35] (exp.)
		1047	0.565	[35] (cal.)
		1044	0.561	[34] (cal.)
		993	–	[84] (exp.)
		993	0.560	[37] (cal.)
		1048	0.565	[36] (cal.)
		1045	0.565	[38] (cal.)
		1048.6	0.556	This work (cal.)
L + $NdNi_3$ ↔ $NdNi_2$	Peritectic	1237	–	[35] (exp.)
		1237	0.649	[35] (cal.)
		1234.8	–	[34] (cal.)
		1213	0.630	[84] (exp.)
		1201	0.647	[37] (cal.)
		1234	0.643	[36] (cal.)
		1237	0.649	[38] (cal.)
		1237.0	0.633	This work (cal.)
L + Nd_2Ni_7 ↔ $NdNi_3$	Peritectic	1341	–	[35] (exp.)
		1341	0.680	[35] (cal.)
		1340.8	–	[34] (cal.)
		1303	0.640	[84] (exp.)
		1286	0.673	[37] (cal.)
		1340	0.673	[36] (cal.)
		1340	0.679	[38] (cal.)
		1341.0	0.665	This work (cal.)
L + $NdNi_5$ ↔ Nd_2Ni_7	Peritectic	1466	–	[35] (exp.)
		1464	0.721	[35] (cal.)
		1465.2	–	[34] (cal.)
		1407	0.660	[84] (exp.)
		1388	0.709	[37] (cal.)
		1465	0.714	[36] (cal.)
		1445	0.714	[38] (cal.)
		1466.7	0.710	This work (cal.)
L ↔ $NdNi_5$	Congruent	1678	0.833	[35] (cal.)
		1677.6	0.833	[34] (cal.)
		1693	0.833	[84] (exp.)
		1689	0.833	[37] (cal.)
		1678	0.833	[36] (cal.)
		1685	0.833	[38] (cal.)
		1677.5	0.833	This work (cal.)
L ↔ Nd_2Ni_{17} + fcc(Ni)	Eutectic	1558	0.918	[35] (cal.)
		1555.2	0.926	[34] (cal.)
		1563	0.910	[84] (exp.)
		1559	0.916	[37] (cal.)
		1569	0.921	[36] (cal.)
		1559	0.917	[38] (cal.)
		1555.1	0.925	This work (cal.)
L + $NdNi_5$ ↔ Nd_2Ni_{17}	Peritectic	1559	0.917	[35] (cal.)
		1577.1	–	[34] (cal.)
		1573	~ 0.910	[84] (exp.)
		1572	0.906	[37] (cal.)
		1572	0.918	[36] (cal.)

Table 8 continued

Reactions	Reaction type	Temperature (K)	Composition (x^L_{Ni})	Reference
		1561	0.915	[38] (cal.)
		1571.2	0.914	This work (cal.)
$Nd_2Ni_{17} \leftrightarrow$ fcc(Ni) + $NdNi_5$	Eutectoid	1550	0.895	[35] (cal.)
		1555	–	[35] (exp.)
		1503.6	–	[34] (cal.)
		1523	–	[84] (exp.)
		1523	0.895	[37] (cal.)
		1524	0.895	[36] (cal.)
		1520.2	0.895	This work (cal.)

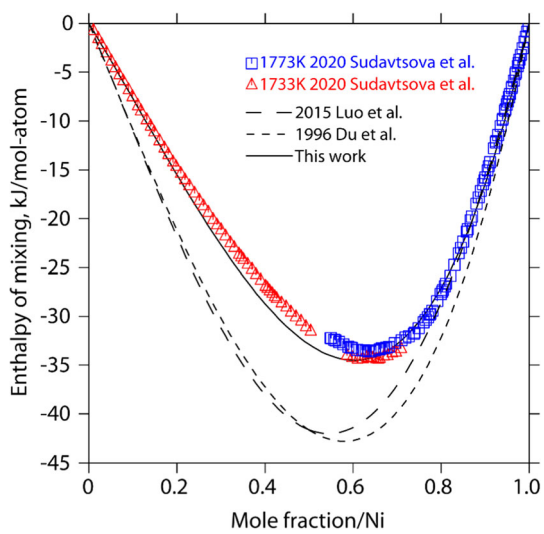


Figure 12 The calculated enthalpy of mixing of the liquid Nd-Ni alloys at 1750 K in comparison with the experimental data [38] and the prior calculation results [36, 37]. (Reference states: liquid Nd and liquid Ni).

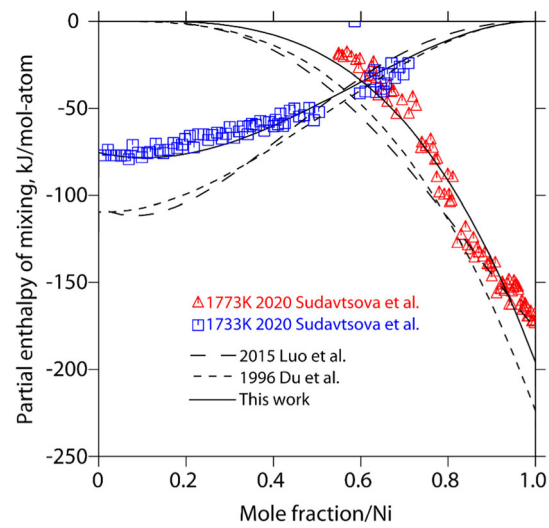


Figure 13 The calculated partial enthalpy of mixing of the liquid Nd-Ni alloys at 1750 K in comparison with the experimental data [38] and the prior calculation results [36, 37]. (Reference states: liquid Nd and liquid Ni).

indicates that the Ho-Ni binary phase diagram needs further experimental investigation.

In Fig. 20, the calculated enthalpy of mixing of the liquid Ho-Ni alloy at 1820 K in this work is shown to be more negative than the calculation results by Kardelass et al. [40]. Figure 21 is the calculated enthalpy of formation of the Ho-Ni intermetallic compounds at 298 K in comparison with the experimental data [82, 88, 89] and the available calculation results [40, 65, 66]. It can be easily seen that the agreement is quite good among all data sets.

Discussion

The rare earth metals (RE)-transition metals (TM) binary systems such as the RE-Mn and RE-Fe binary systems [47–50] exhibit a rather systematic trend for

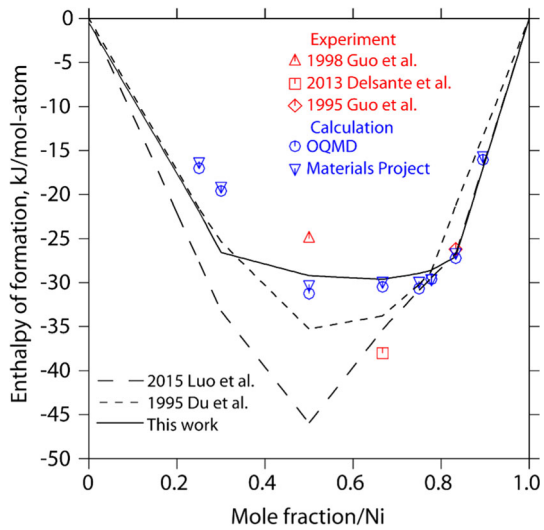


Figure 14 The calculated enthalpy of formation of the Nd-Ni intermetallic compounds at 298 K in comparison with the experimental data [82, 124, 125] and the prior calculation results [36, 37, 65, 66]. (Reference states: dhcp-Nd and fcc-Ni).

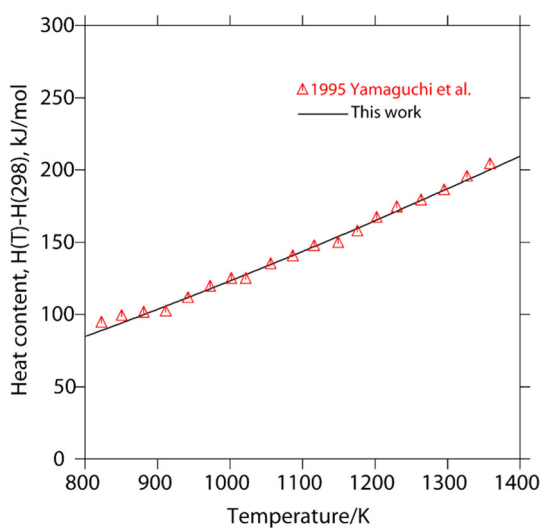


Figure 15 The calculated heat content of $NdNi_5$ in comparison with the experimental data [67].

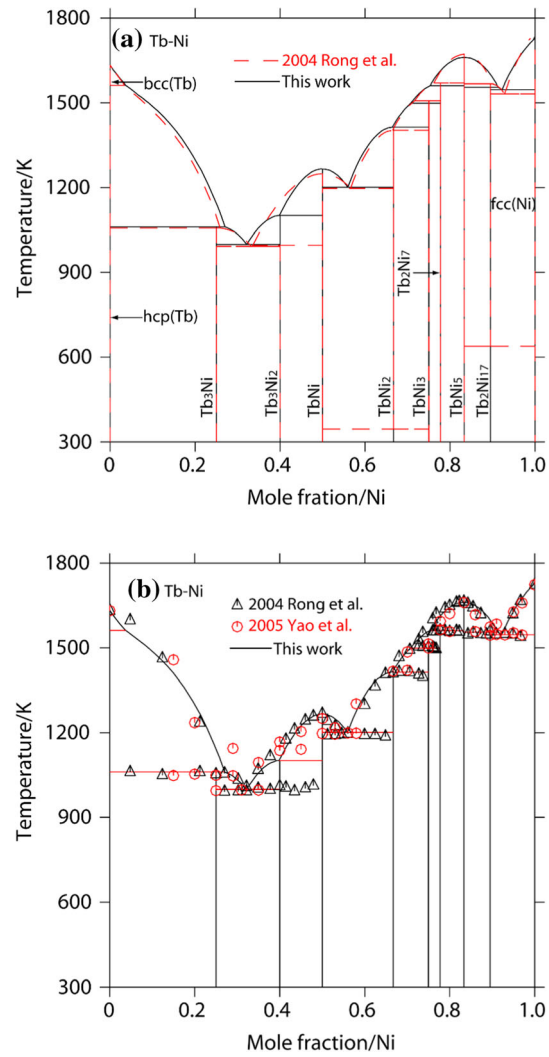


Figure 16 The calculated Tb-Ni binary phase diagram in comparison with the experimental data [39, 86] and the prior calculation results [86].

the dependency of the phase equilibria and thermodynamic properties on the atomic number of RE because the rare earth elements systematically change in their physical and chemical properties due to their unique electronic configurations. In the following section, the presently calculated results of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems are examined together with the results of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary systems reported in Refs. [90–95] to see if a similar trend exists.

Phase equilibria

The RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary phase diagrams were calculated with the thermodynamic descriptions available in Refs. [90–95] and are shown in Fig. 22. As can be seen, the Pr-Ni binary system has seven stable intermetallic compounds, $PrNi_5$, Pr_2Ni_7, $PrNi_3$, $PrNi_2$, PrNi, Pr_7Ni_3 and Pr_3Ni

Table 9 Invariant reactions in the Tb-Ni binary system

Reactions	Reaction type	Temperature (K)	Composition (x^L_{Ni})	Reference
L + bcc(Tb) ↔ hcp(Tb)	Peritectic	1562	0.034	[86] (cal.)
		1562.0	0.037	This work (cal.)
L + hcp(Tb) ↔ Tb_3Ni	Peritectic	1056	0.259	[86] (cal.)
		1055	–	[86] (exp.)
		1062	–	[39] (exp.)
		1060.3	0.271	This work (cal.)
L ↔ Tb_3Ni_2 + Tb_3Ni	Eutectic	992	0.332	[86] (cal.)
		991	–	[86] (exp.)
		996	–	[39] (exp.)
		999.0	0.321	This work (cal.)
L + TbNi ↔ Tb_3Ni_2	Peritectic	996	0.335	[86] (cal.)
		1002	–	[86] (exp.)
		~ 1104	–	[39] (exp.)
		1101.5	0.399	This work (cal.)
L ↔ TbNi	Congruent	1248	0.500	[86] (cal.)
		1266	0.500	[86] (exp.)
		1250	0.500	[39] (exp.)
		1266.0	0.500	This work (cal.)
L ↔ TbNi + $TbNi_2$	Eutectic	1196	0.566	[86] (cal.)
		1191	–	[86] (exp.)
		1199	–	[39] (exp.)
		1201.0	0.559	This work (cal.)
L + $TbNi_3$ ↔ $TbNi_2$	Peritectic	1403	0.664	[86] (cal.)
		1407	–	[86] (exp.)
		1417	–	[39] (exp.)
		1414.3	0.664	This work (cal.)
L + Tb_2Ni_7 ↔ $TbNi_3$	Peritectic	1507	0.716	[86] (cal.)
		1500	–	[86] (exp.)
		1499	–	[39] (exp.)
		1498.3	0.707	This work (cal.)
L + $TbNi_5$ ↔ Tb_2Ni_7	Peritectic	1569	0.761	[86] (cal.)
		1558	–	[86] (exp.)
		1560.8	0.753	This work (cal.)
L ↔ $TbNi_5$	Congruent	1672	0.833	[86] (cal.)
		1663	0.833	[86] (exp.)
		1660	0.833	[39] (exp.)
		1660.3	0.833	This work (cal.)
L + $TbNi_5$ ↔ Tb_2Ni_{17}	Peritectic	1568	0.889	[86] (cal.)
		1550	–	[86] (exp.)
		1557	–	[39] (exp.)
		1554.8	0.913	This work (cal.)
L ↔ fcc(Ni) + Tb_2Ni_{17}	Eutectic	1532	0.928	[86] (cal.)
		1545	–	[86] (exp.)
		1544	–	[39] (exp.)
		1547.0	0.922	This work (cal.)

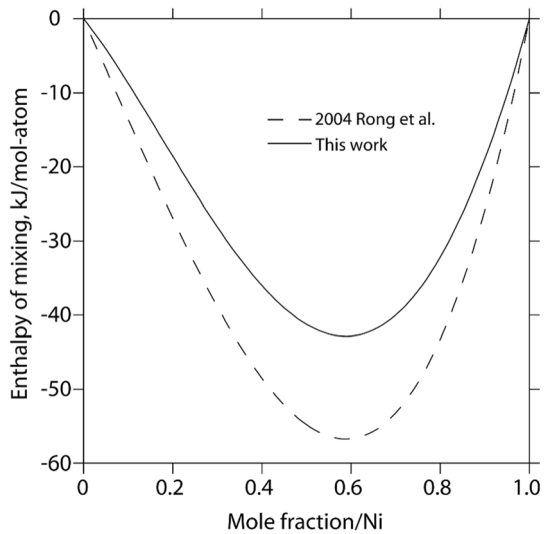


Figure 17 The calculated enthalpy of mixing of the liquid Tb-Ni alloys at 1820 K in comparison with the prior calculation results [86]. (Reference states: liquid Tb and liquid Ni).

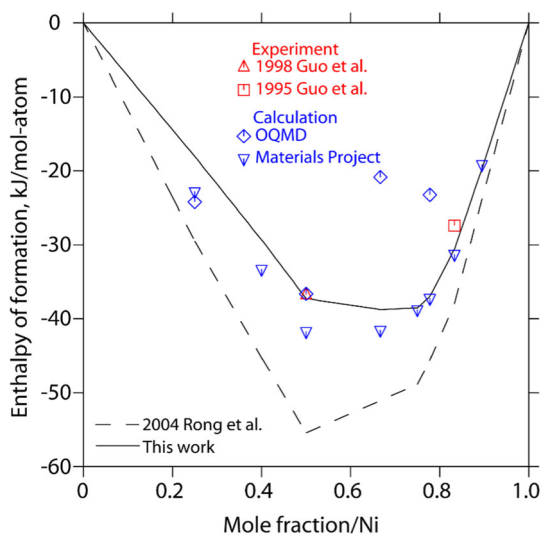


Figure 18 The calculated enthalpy of formation of the Tb-Ni intermetallic compounds at 298 K in comparison with the experimental data [82, 87] and the prior calculation results [65, 66, 86]. (Reference states: hcp-Tb and fcc-Ni).

according to the experimental results determined by Vogel and Fülling [54], Qi et al. [51], Lemaire and Paccard [96], and Pan and Cheng [97]. This system was thermodynamically assessed by Du et al. [91], Huang and Lograsso [98], and Rahou et al. [90]. The Sm-Ni binary system was investigated experimentally by Pan and Zheng [99], Pan and Nash [100] and Borzone et al. [101] and was assessed thermodynamically by Su et al. [102], and Rahou and Mahdouk

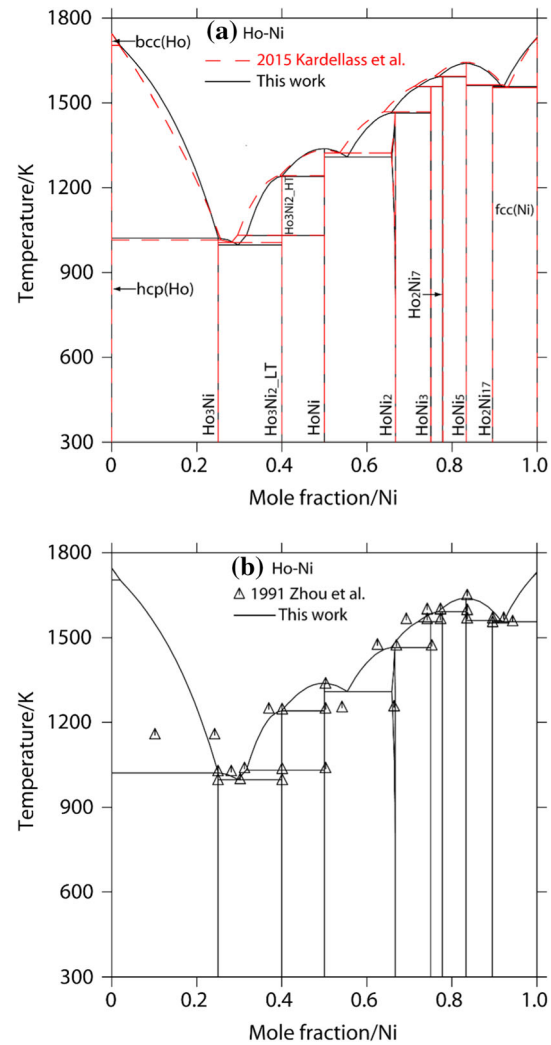


Figure 19 The calculated Ho-Ni binary phase diagram in comparison with the experimental data [45] and the prior calculation results [40].

[92]. Ten intermetallic compounds, Sm_3Ni, Sm_7Ni_3, Sm_3Ni_2, SmNi, $SmNi_2$, $SmNi_3$, Sm_2Ni_7, Sm_5Ni_{19}, $SmNi_5$ and Sm_2Ni_{17}, are stable. In the Gd-Ni binary system, there are seven intermetallic compounds (e.g., Gd_3Ni, GdNi, $GdNi_2$, $GdNi_3$, Gd_2Ni_7, $GdNi_5$ and Gd_2Ni_{17}) according to the experimental results determined by Spedding and Daane [103], Novy et al. [104], Copeland and Kato [105], Pan et al. [106], Pan and Nash [107], Dischinger and Schaller [108], and Xu et al. [109], and were assessed thermodynamically by Xia and Jin [110], and Rahou and Mahdouk [93]. The Dy-Ni binary system was studied experimentally by Dwight et al. [111], Baenziger et al. [112], Buschow and Goot [113], and Zheng and Wang [114], and there are ten intermetallic compounds, Dy_3Ni, Dy_3Ni_2,

Table 10 Invariant reactions in the Ho-Ni binary system

Reactions	Reaction type	Temperature (K)	Composition (x_{Ni}^{L})	Reference
L + bcc(Ho) ↔ hcp(Ho)	Peritectic	1703.0	0.020	This work (cal.)
L + hcp(Ho) ↔ Ho_3Ni	Peritectic	1014	0.256	[40] (cal.)
		1023	0.300	[45] (exp.)
		1020.2	0.251	This work (cal.)
L ↔ Ho_3Ni + Ho_3Ni_2_LT	Eutectic	1006	0.281	[40] (cal.)
		993	–	[45] (exp.)
		996.3	0.297	This work (cal.)
Ho_3Ni_2_HT ↔ Ho_3Ni_2_LT	Polymorphic	1031	0.400	[40] (cal.)
		1033	0.400	[45] (exp.)
		1030.2	0.400	This work (cal.)
L + HoNi ↔ Ho_3Ni_2_HT	Peritectic	1241	0.394	[40] (cal.)
		1243	–	[45] (exp.)
		1240.4	0.399	This work (cal.)
L ↔ HoNi	Congruent	1333	0.500	[40] (cal.)
		1333	0.500	[45] (exp.)
		1338.0	0.500	This work (cal.)
L ↔ HoNi + $HoNi_2$	Eutectic	1322	0.536	[40] (cal.)
		1248	0.540	[45] (exp.)
		1308.9	0.553	This work (cal.)
L + $HoNi_3$ ↔ $HoNi_2$	Peritectic	1468	0.640	[40] (cal.)
		1468	–	[45] (exp.)
		1465.2	0.657	This work (cal.)
L + Ho_2Ni_7 ↔ $HoNi_3$	Peritectic	1558	0.712	[40] (cal.)
		1558	–	[45] (exp.)
		1555.8	0.720	This work (cal.)
L + $HoNi_5$ ↔ Ho_2Ni_7	Peritectic	1593	0.759	[40] (cal.)
		1593	–	[45] (exp.)
		1590.9	0.768	This work (cal.)
L ↔ $HoNi_5$	Congruent	1643	0.833	[40] (cal.)
		1643	0.833	[45] (exp.)
		1638.2	0.833	This work (cal.)
L + $HoNi_5$ ↔ Ho_2Ni_{17}	Peritectic	1563	0.919	[40] (cal.)
		1563	–	[45] (exp.)
		1560.3	0.915	This work (cal.)
L ↔ fcc(Ni) + Ho_2Ni_{17}	Eutectic	1553	0.931	[40] (cal.)
		1553	0.930	[45] (exp.)
		1556.0	0.921	This work (cal.)

DyNi, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, $DyNi_4$, Dy_4Ni_{17}, $DyNi_5$ and Dy_2Ni_{17}. Thermodynamic calculation of the Dy-Ni binary system was carried out by Li and Han [94]. The thermodynamic assessment of the Er-Ni binary system was performed by Du et al. [91] based on the experimental results measured by Buschow [115], Moreau et al. [116], and Pan and Nash [117]. There are eleven intermetallic compounds, Er_3Ni, Er_3Ni_2, ErNi, $ErNi_2$, $ErNi_3$, Er_2Ni_7, $ErNi_4$, Er_4Ni_{17}, Er_5Ni_{22}, $ErNi_5$ and Er_2Ni_{17} in this system. The Y-Ni binary system was measured by Beaudry and Daane [118] and Domagala et al. [119] and was assessed thermodynamically by Mezbahul-Islam and Medraj [120], Nash [121], Mattern et al. [122] and Du and Zhang [95]. Nine intermetallic compounds, Y_3Ni, Y_3Ni_2, YNi, YNi_2, YNi_3, Y_2Ni_7, YNi_4, YNi_5 and Y_2Ni_{17}, are stable.

Combining the calculated results of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary systems mentioned above and the calculated results of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems in this work, Fig. 23 and Table 11 show the

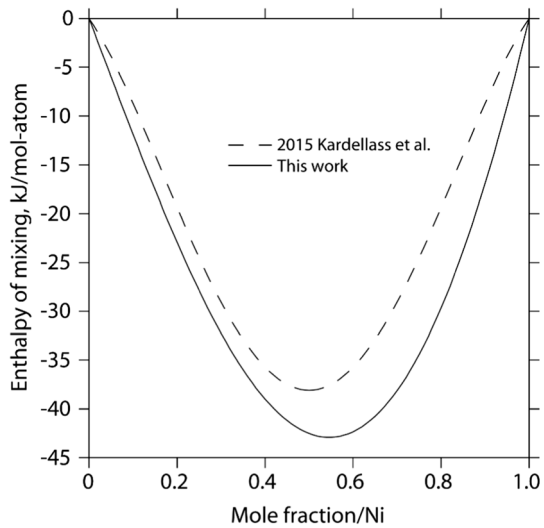


Figure 20 The calculated enthalpy of mixing of the liquid Ho-Ni alloys at 1820 K in comparison with the prior calculation results [40]. (Reference states: liquid Ho and liquid Ni).

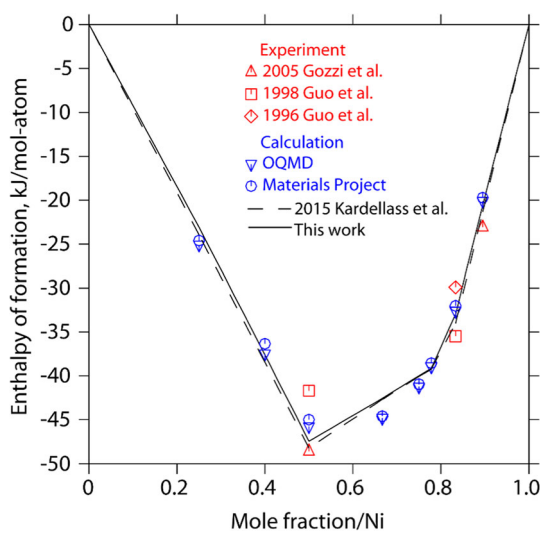


Figure 21 The calculated enthalpy of formation of the Ho-Ni intermetallic compounds at 298 K in comparison with the experimental data [82, 88, 89] and the prior calculation results [40, 65, 66]. (Reference states: hcp-Ho and fcc-Ni).

temperatures and types of the invariant reactions for the formation of intermetallic compounds in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems. There are four stable intermetallic compounds RENi, $RENi_3$, RE_2Ni_7 and $RENi_5$ in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems in Table 11. As shown in Fig. 23a, RE_7Ni_3 exists only in the RE-Ni (RE = La, Ce, Pr, Nd and Sm) binary systems, while RE_3Ni_2 exists only in the RE-Ni (RE = Sm, Y, Tb, Dy, Ho and Er) binary systems. In addition, RE_3Ni is only not stable in the Ce-Ni binary system, while $RENi_2$ is only not stable in the La-Ni binary system. It is interesting that the La-Ni binary system shows additional RE_2Ni_3 and RE_7Ni_{16}, while the Er-Ni binary system has the additional RE_5Ni_{22}. On the other hand, the RE-Ni (RE = Sm, Dy and Er) binary systems have additional $RENi_4$ and the RE-Ni (RE = La, Ce and Sm) binary systems have additional RE_5Ni_{19}. In particular, RE_4Ni_{17} only exists in the Dy-Ni and Er-Ni binary systems, and RE_2Ni_{17} exists in the RE-Ni (RE = Nd, Sm, Y, Gd, Tb, Dy, Ho and Er) binary systems.

It was found that RE_7Ni_3 is formed by the congruent reaction in the La-Ni, Ce-Ni and Pr-Ni binary systems and is formed by the peritectic reaction in the Nd-Ni and Sm-Ni binary systems in Table 11. Moreover, RE_3Ni is formed by the congruent reaction in the RE-Ni (RE = La, Pr, Nd and Sm) binary systems, but by peritectic reaction in the RE-Ni (RE = Y, Gd, Tb, Dy, Ho and Er) binary systems. RENi and $RENi_5$ are formed by the congruent reaction. In addition, $RENi_2$, $RENi_3$ and RE_2Ni_7 are formed by the peritectic reaction. It is worth noting that the intermetallic compounds (e.g., $RENi_2$, $RENi_3$, RE_2Ni_7) between RENi and $RENi_5$ are formed by the peritectic reaction in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Y, Gd, Tb, Dy, Ho and Er) binary systems, except for Er_2Ni_7 which is formed by the peritectoid reaction. RE_2Ni_{17} was formed by the peritectic reaction of $L + RENi_5 \leftrightarrow RE_2Ni_{17}$. According to the reported systematic trends in the RE-Fe and RE-Mn binary systems [47–50], the reaction temperatures of the RE-Ni intermetallic compounds increased dependent on the increase of atomic numbers of rare earth metals. In the RE-Ni binary system, the melting points of pure rare earth metals and the reaction temperatures of intermetallic compounds also show the same tendency in this work. Among them, $RENi_3$ (RE = La,

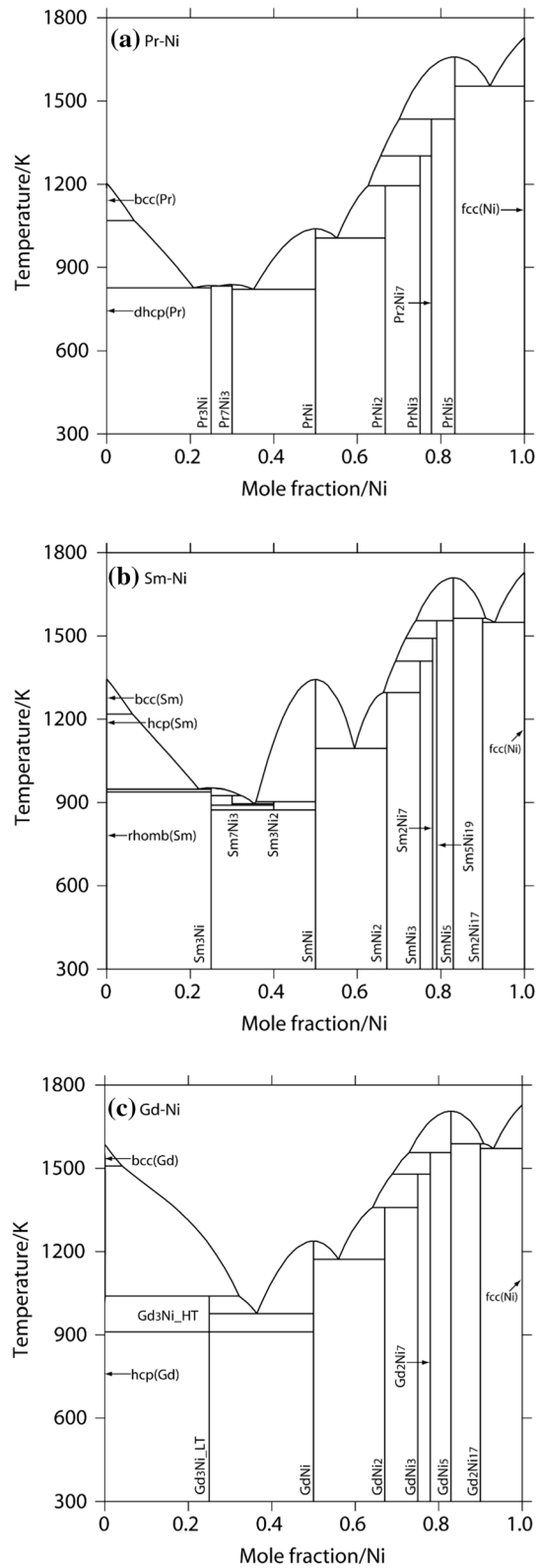


Figure 22 The calculated RE-Ni (RE = Gd, Tb, Dy, Ho, Er and Y) binary phase diagrams in the reported literature. **a** Pr-Ni [90], **b** Sm-Ni [92], **c** Gd-Ni [93], **d** Dy-Ni [94], **e** Er-Ni [91], **f** Y-Ni [95].

Pr, Nd, Sm, Gd, Tb, Dy, Ho, and Er) is strictly in agreement with the trend that the higher the atomic number of rare earth elements, the higher the temperature in Fig. 23b. In general, other intermetallic compounds (e.g., RE_3Ni, RE_7Ni_3, RE_3Ni_2, RENi, $RENi_2$, RE_2Ni_7) also show this trend in Fig. 23a–c, but this trend is not clear for $RENi_5$ and RE_2Ni_{17}.

Thermodynamic properties

Figure 24 shows the calculated enthalpies of mixing of liquid RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) alloys. The calculated enthalpies of mixing of liquid RE-Ni (RE = La, Pr, Nd, Gd, Tb, Ho) alloys display the tendency that the enthalpies of mixing of liquid phase become much more negative with the higher number the atomic number of the rare earth metals. In general, the most negative values of the enthalpies of mixing of liquid RE-Ni alloys are at around 60 at.% Ni (between 50 and 70 at.% Ni). In this work, this systematic trend in thermodynamic properties of the RE-Ni binary systems was taken into account during the optimization of the Tb-Ni and Ho-Ni binary systems due to the limited experimental information. It is noted that the most negative values of the mixing enthalpies of liquid Sm-Ni alloys are more negative than those in the RE-Ni (RE = Gd, Tb, Ho, Er and Y) binary systems, and the most negative values of the mixing enthalpies of liquid Dy-Ni alloys are the most negative in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems. However, due to the lack of experimental information on the enthalpies of mixing of liquid RE-Ni (RE = Sm, Tb, Dy, Ho and Er) alloys, it indicates that further experiments still need to measure the mixing enthalpies of liquid RE-Ni alloys.

Figure 25 shows the excess entropy of mixing of liquid RE-Ni (RE = La, Ce, Nd, Tb and Ho) alloys calculated in this work and the excess entropy of mixing of liquid RE-Ni (RE = Pr, Sm, Y, Gd, Dy and Er) alloys in Refs. [90–95]. Okamoto [123] indicated that the excess entropy of mixing of a liquid phase could not have a very large positive value and

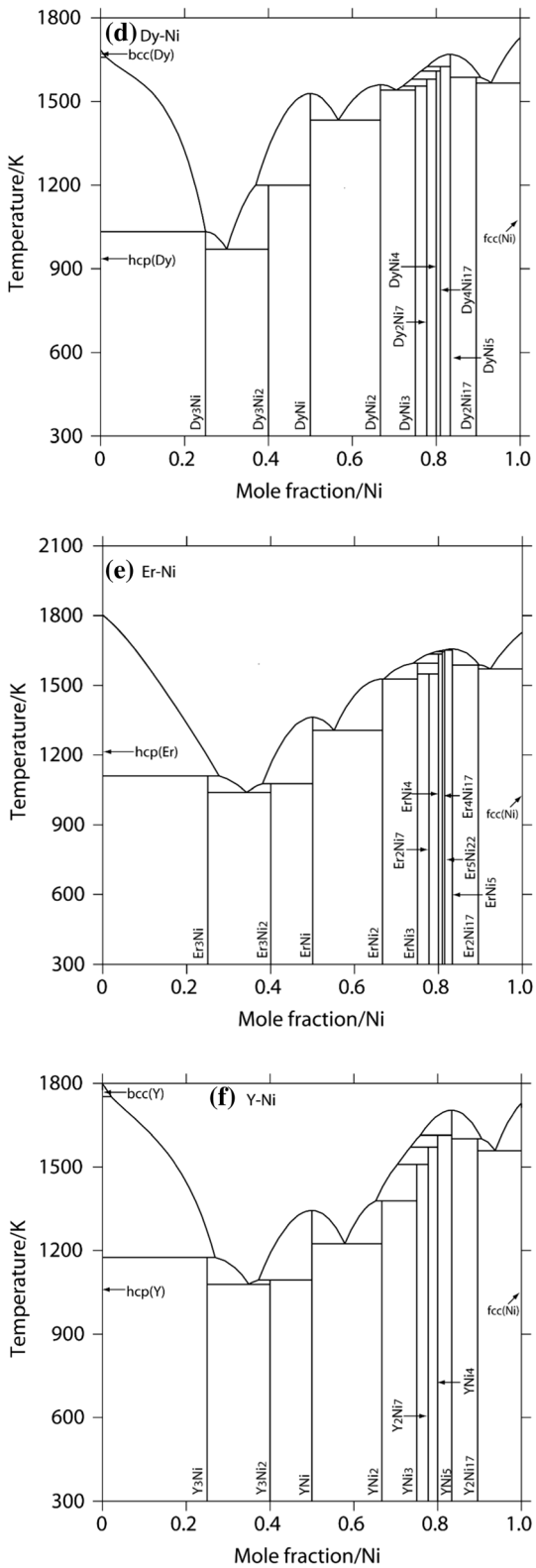


Figure 22 continued.

suggested the excess entropy of mixing of liquid metallic alloys would be expected to be in the range between -10 and 5 J/mol. The value of excess entropy of mixing of liquid RE-Ni (RE = La, Ce, Nd, Tb and Ho) alloys calculated in this work is between -10 and 5 J/mol, which is in good agreement with the suggestions of Okamoto [123]. In general, the most negative value of excess entropy of mixing of liquid RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Ho and Er) alloys is between 60 and 80 at.% Ni. In particular, there is obvious positive value at around 20 at.% Ni in the RE-Ni (RE = La, Ce, Pr and Sm) systems.

As shown in Fig. 26, the calculated enthalpies of formation of the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) intermetallic compounds (e.g., RE_3Ni, RE_7Ni_3, RE_3Ni_2, RENi, $RENi_2$, RE_2Ni_7, $RENi_5$ and RE_2Ni_{17}) at room temperature are located within the range of − 10 and − 50 kJ/mol-atom. The values of enthalpies of formation of the RE-Ni intermetallic compounds which are formed between 50 and 80 at.% Ni are the most negative and show the tendency that the enthalpies of formation in the RE-Ni binary systems become much more negative at room temperature with the increase of the atomic number of rare earth metals. It was found that the calculated enthalpies of formation of the Sm-Ni intermetallic compounds at room temperature are little more negative than those of the Gd-Ni intermetallic compounds using the CALPHAD method. At present, the experimental data of the enthalpy of formation of the RE-Ni (RE = Gd, Tb, Dy, Ho and Er) intermetallic compounds at room temperature are relatively limited, and the results measured experimentally and calculated theoretically are not quite consistent with each other. It means that the enthalpies of formation of the RE-Ni intermetallic compounds need further experimental measurement. In addition, the enthalpies of formation of the RE-Ni intermetallic compounds in the RE-Ni binary systems are much more negative in the RE-rich part, resulting in higher phase stability of these intermetallic compounds. According to the systematic analysis of the enthalpies of mixing of liquid RE-Ni alloys and the enthalpies of formation of the RE-Ni intermetallic compounds, the values are generally much more negative as the atomic number of the rare earth metals increases, which is similar with the tendency in the RE-Fe and RE-Mn binary systems [47–50].

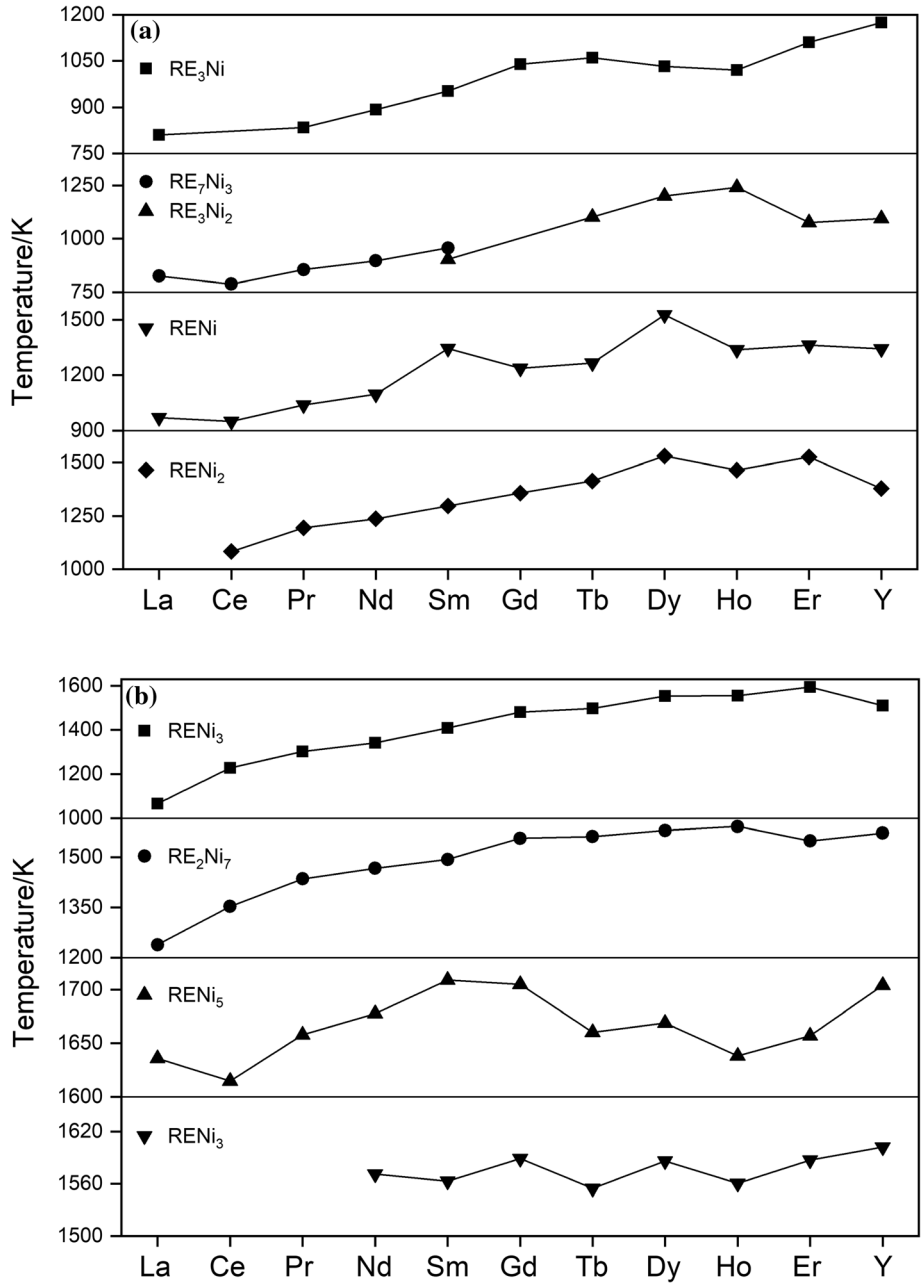


Figure 23 The reaction temperatures of intermetallic compounds in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems.

Table 11 Temperatures of formation reaction for intermetallic compounds in the RE-Ni binary systems

Systems	La-Ni	Ce-Ni	Pr-Ni	Nd-Ni	Sm-Ni	Gd-Ni	Tb-Ni	Dy-Ni	Ho-Ni	Er-Ni	Y-Ni
RE_3Ni	810.9^C	–	835^C	892.9^C	952^C	1039^P	1060.3^P	1032^P	1020.2^P	1110^P	1175^P
RE_7Ni_3	814.0^C	781.8^C	839^C	873.8^P	925^P	–	–	–	–	–	–
RE_3Ni_2	–	–	–	–	903^P	–	1101.5^P	1200^P	1240.4^P	1076^P	1094^P
RENi	969.6^C	950.1^C	1039^C	10968^C	1345^C	1238^C	1266.0^C	1527^C	1338.0^C	1363^C	1343^C
RE_2Ni_3	944.1^P	–	–	–	–	–	–	–	–	–	–
$RENi_2$	–	1083.0^P	1194^P	1237.0^P	1297^P	1358^P	1414.3^P	1531^P	1465.2^P	1527^P	1379^P
RE_7Ni_{16}	968.2^P	–	–	–	–	–	–	–	–	–	–
$RENi_3$	1066.2^P	1228.3^P	1302^P	1341.0^P	1409^P	1481^P	1498.3^P	1554^P	1555.8^P	1595^P	1510^P
RE_2Ni_7	1238.4^P	1352.9^P	1435^P	1466.7^P	1493^P	1556^P	1560.8^P	1579^P	1590.9^P	1548^{Pd}	1571^P
$RENi_4$	–	–	–	–	–	–	–	1608^P	–	1634^P	1614^P
RE_4Ni_{17}	–	–	–	–	–	–	–	1625^P	–	1644^P	–
RE_5Ni_{19}	1277.7^P	1394.1^P	–	–	1555^P	–	–	–	–	–	–
RE_5Ni_{22}	–	–	–	–	–	–	–	–	–	1651^P	–
$RENi_5$	1635.9^C	1614.7^C	1658^C	1677.5^C	1709^C	1705^C	1660.3^C	1669^C	1638.2^C	1657^C	1704^C
RE_2Ni_{17}	–	–	–	1571.2^P	1563^P	1589^P	1554.8^P	1586^P	1560.3^P	1587^P	1602^P

C: congruent, P: peritectic, Pd: peritectoid

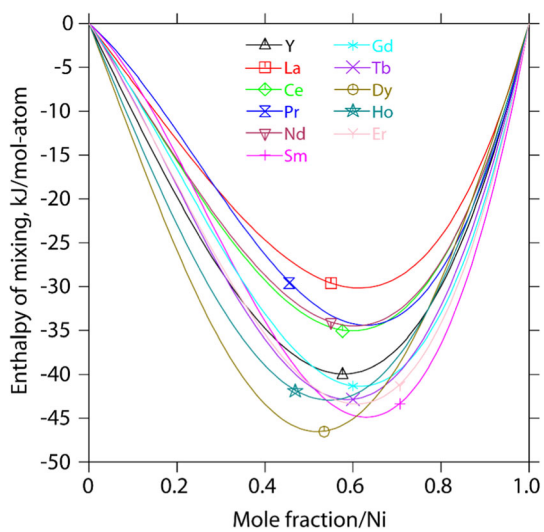


Figure 24 The calculated enthalpy of mixing of liquid RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) alloys. (Reference states: liquid RE and liquid Ni).

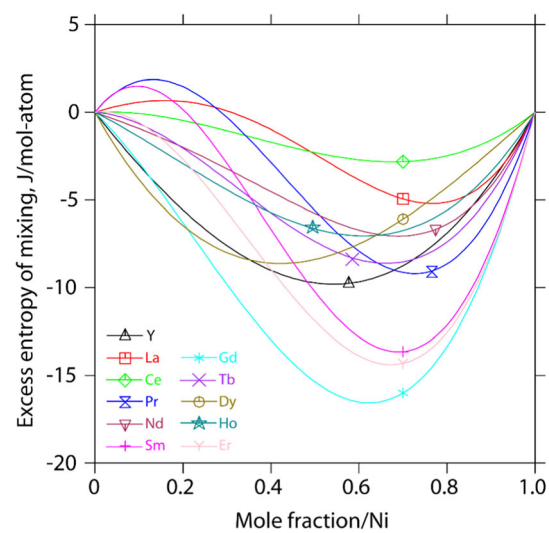
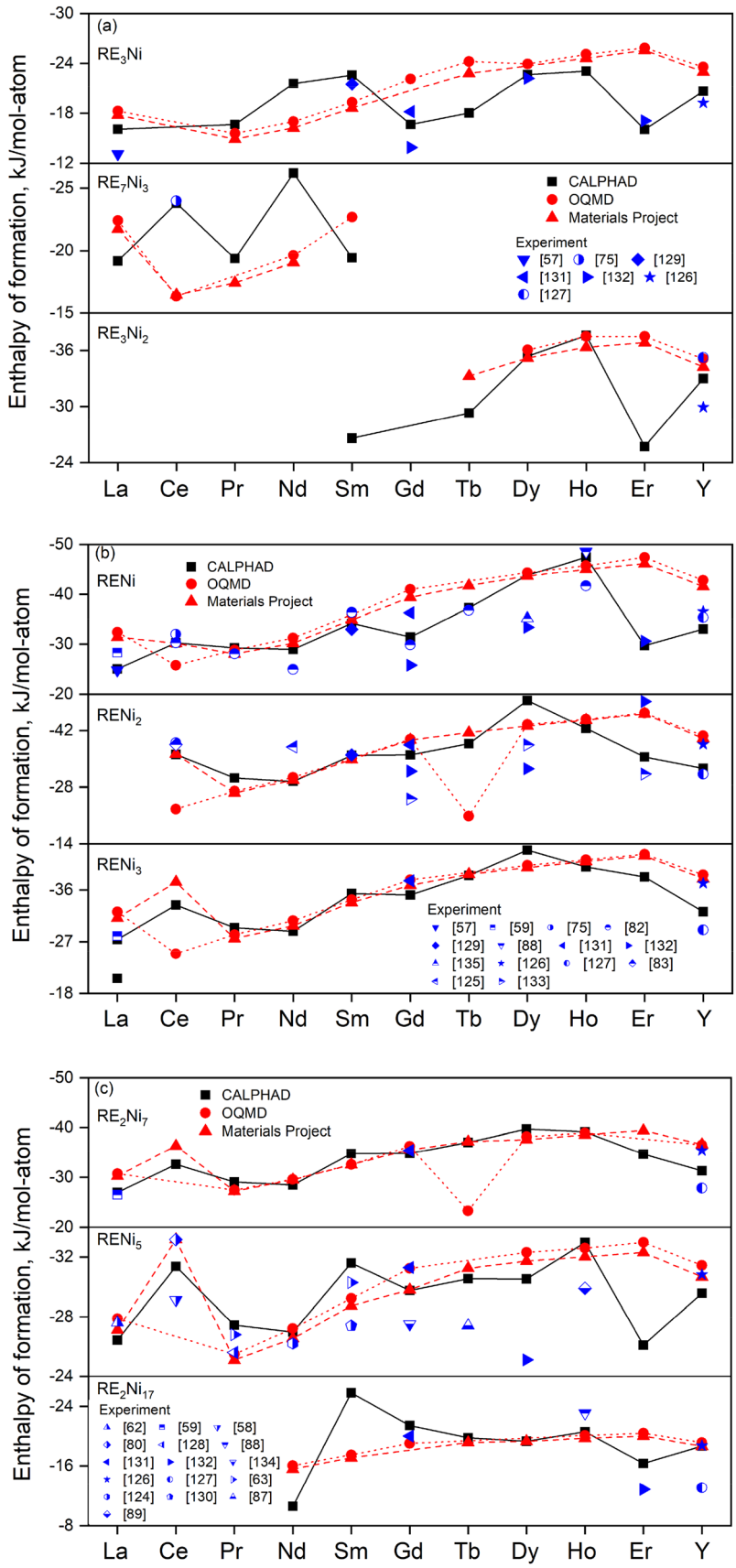


Figure 25 The calculated excess entropy of mixing of liquid phase in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems.

Conclusions

The RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems were assessed by the CALPHAD method on the basis of the reported experimental results. A set of self-consistent thermodynamic parameters of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems was obtained. The calculated phase diagrams and thermodynamic properties of the RE-Ni (RE = La, Ce, Nd, Tb and Ho) binary systems are in good agreement with the experimental data and the reported results. In combination with the reported assessments of the RE-Ni (RE = Pr, Sm, Gd, Dy, Er and Y) binary systems, the phase equilibria and thermodynamic


(a)
RE3Ni
RE7Ni3
RE3Ni2
CALPHAD
OQMD
Materials Project
Experiment
[57] [75] [129]
[131] [132] [126]
[127]
(b)
RENi
RENi2
RENi3
Experiment
[57] [59] [75] [82]
[129] [88] [131] [132]
[135] [126] [127] [83]
[125] [133]
(c)
RE2Ni7
RENi5
RE2Ni17
Experiment
[62] [59] [58]
[80] [128] [88]
[131] [132] [134]
[126] [127] [63]
[124] [130] [87]
[89]
Enthalpy of formation, kJ/mol-atom
La Ce Pr Nd Sm Gd Tb Dy Ho Er Y

◀**Figure 26** The calculated enthalpy of formation of intermetallic compounds at 298 K with the experimental data [57–59, 62, 63, 75, 80, 82, 83, 87–89, 124–135] in the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems. (Reference states: solid RE and solid Ni).

properties of the RE-Ni (RE = La, Ce, Pr, Nd, Sm, Gd, Tb, Dy, Ho, Er and Y) binary systems were analyzed systematically. In general, the larger the atomic number of RE is, the higher the phase transition temperatures of the RE-Ni intermetallic compounds are. It was found that the enthalpies of mixing of the liquid RE-Ni alloys and enthalpies of formation of the RE-Ni intermetallic compounds are generally much more negative with the increase of the atomic number of RE, which is similar with the trend observed in the RE-Fe and RE-Mn binary systems.

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Declarations

Conflict of interest The authors of the manuscript 'Phase equilibria and thermodynamic properties in the RE-Ni (RE = rare earth metals) binary systems' declare no conflict of interest.

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